

Highlights

A Cartesian multipole hierarchy for the voxel: closed-form single-site T -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes

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- A Cartesian multipole hierarchy gives the cube's single-site T -matrix in closed form
- A voxel needs two bases: one for the medium and one for the wavefield
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T -matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

A Cartesian multipole hierarchy for the voxel: closed-form single-site T -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes

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Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$, with leading error $c_p(kd)^{2p+2}$ in closed form; the orders hold for SV and SH waves and at oblique incidence. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$: the two bases must be matched, and the error is known from the medium before anything is solved. In a plane-stratified background the orders persist: the background changes the field the voxels must represent, not how well they represent it. The single site is the first member of a

Cartesian multipole hierarchy, an elastodynamic extension of the equivalent inclusion method, whose coefficients are moments of the Green's tensor over the cube, in closed form on four constants. Its blocks gain accuracy in pairs: with third gradients it is 300 times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order on a graded sphere.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background,

16 which the model prescribes. The other is the *wavefield*: the displacement and
 17 strain inside the voxel, which the calculation must find. Each is expanded
 18 in a basis on the cell, and the two bases are separate choices. Here both
 19 are Legendre polynomials, the wavefield to degree p and the contrast to
 20 degree r ; the usual scheme, a uniform field in a cell of constant properties,
 21 is $p = r = 0$. Neither choice is new in itself. In higher-order volume integral
 22 methods for electromagnetics the material within an element is represented
 23 by polynomials whose orders are independent of those of the field [1], as it
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What
 25 has been missing is the law that relates the two bases: the order a given pair
 26 attains, and the error it leaves. The answer decides the cost of everything
 27 built on it, because a voxel that represents its contents more accurately can
 28 be larger, and the number of voxels, not the size of each one's state, sets the
 29 cost of the multiple-scattering solve.

30 *Why the question is hard to answer.* For the media such a scheme is built for,
 31 there is no exact solution. Its error cannot be measured against the answer.
 32 Comparison with another numerical method mixes the errors of both. And a
 33 refinement study can mislead, because a voxel scheme can converge smoothly
 34 to the wrong limit. §4 shows how that happens with the most natural choice
 35 of coupling, the Green's tensor evaluated between voxel centres: it carries
 36 a local term whose size per unit volume does not shrink as the voxels do,
 37 and unless the single voxel's response is matched to cancel it, refinement
 38 converges to the wrong answer without revealing it.

39 *The method: a layer of voxels.* This paper therefore answers the question
 40 in the simplest configuration in which a voxel scheme runs in full (every

41 voxel, every coupling, the whole multiple-scattering series, at normal and
 42 at oblique incidence) and the exact answer is nevertheless known: a layer
 43 filled with voxels. Because cubes tile space, a layer of identical voxels does
 44 not approximate a homogeneous layer; it *is* one. Any difference between
 45 its reflection and transmission and those of the continuous layer is therefore
 46 error of the discretisation alone, with no modelling error and no shape error
 47 behind which to hide. A sphere, the usual test object for volume-integral and
 48 discrete-dipole methods [3], cannot give this: its error mixes the discretisation
 49 with the staircase approximation of a curved surface by cubes. The layer
 50 plays the part of the patch test in finite elements: reproducing a uniform
 51 medium correctly is necessary, though not sufficient, for a scheme to be
 52 trusted in a heterogeneous one. The experiment is then made harder in
 53 three steps, each with an exact solution of its own: a randomly stratified
 54 layer, in which every cell is homogeneous but the cells differ; a layer whose
 55 properties vary smoothly within every cell, which is where the basis for the
 56 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 57 surface, which carries the layer's prediction into three dimensions without a
 58 staircase. The coupling and the single site that the layer validates are the
 59 same in any medium; a laterally heterogeneous medium remains to be tested.

60 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges
 61 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 62 each cell therefore caps every voxel at second order in a medium that varies
 63 within the cells, however rich the basis of its field; and a contrast of higher
 64 degree than the field is wasted. (2) *The leading error is in closed form, for*
 65 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$

66 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the orders 2, 4 and 6 of
 67 $p = 0, 1$ and 2 hold for P, SV and SH waves at normal and oblique incidence.
 68 For the collocation scheme, the uniform-field voxel in its simplest form, the
 69 error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§7), and the
 70 opposite signs explain why no correction of the single site can improve both
 71 (§12). (3) *The error that remains is the single site's, and it is the projection*
 72 *error of the medium.* The single-site T -matrix sums the rescattering within a
 73 voxel to all orders and the coupling between voxels is exact, but the T -matrix
 74 is that of the voxel as represented, its contrast projected to degree r and its
 75 field held to degree p . The error of the term of second order in the contrast
 76 equals the *relative projection error* of the medium, the fraction of the squared
 77 norm of its profile that polynomials of degree $\min(p, r)$ on the cells cannot
 78 represent (Eq. (43)): exactly in the long-wave limit, and to within 1.3% for
 79 all nine pairs with $p, r \leq 2$ (§9). The wavefield basis must therefore carry
 80 what the medium imprints on the field, not only the wave, and the two bases
 81 must be matched. (4) *The error is known in advance.* The projection error
 82 is a property of the medium and the grid alone, so the cell size and the
 83 degree that a required accuracy needs are chosen before anything is solved.
 84 (5) *Where the medium is piecewise constant on the cells the projection error*
 85 *vanishes*, a constant suffices for the medium, and the degree of the field
 86 alone sets the order: in a randomly stratified model one first-moment voxel
 87 per model layer reaches a reflection accuracy of 10^{-6} , where the uniform-field
 88 voxel needs about forty (§9), and in the uniform layer it allows voxels 58 times
 89 larger in each direction (Appendix Appendix I). (6) *In three dimensions* the
 90 smoothly graded sphere converges at second order with the mean-only voxel,

91 as the layer predicts (§10), and at fourth order with the first-moment voxel
 92 (§11).

93 *What makes the experiment exact.* The answer above can be trusted only if
 94 nothing but the two bases is approximated. Three results, set against the
 95 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 96 The integrated Green tensor and the filtered coupled dipoles approximate the
 97 field of a cell; for cubes the source-cell average is exact, because the cube's
 98 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 99 at normal incidence the lattice kernel is in closed form with no lattice sum.
 100 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 101 coupled-dipole method, is here a derived object (§6): the moments of the
 102 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 103 without adding it by hand, the cube's departure from the sphere as a pure
 104 cubic anisotropy, and the closure's error against exact scattering by a sphere
 105 as a real $(ka)^2$ term in closed form. The self term the single site carries is the
 106 one the averaged kernel omits, so the discrete system is a collocation of the
 107 continuum equation. (c) *The elastic state.* A voxel carries a force (through
 108 its density contrast) and a moment (through its stiffness contrast), so its
 109 state has nine components, displacement and strain, the coupling is a 9×9
 110 tensor, and the single site's self-interaction is the elastic depolarisation of a
 111 cube, the discrete counterpart of the inclusion tensor of Eshelby [4]. Cubes
 112 serve all three: they tile space, so a uniform region cut into cubes is still that
 113 region, and their regular lattice makes the coupling a convolution, solved by
 114 fast transforms.

115 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
 116 small scatterers that interact through the background Green's function is the
 117 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics
 118 it became a numerical method as the coupled-dipole method, or discrete
 119 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
 120 dipole driven by the incident field and by the fields of every other dipole
 121 [7, 8]. The method is a collocation of the volume integral equation for the
 122 electric field, with the cell's self-interaction absorbed into its polarisability
 123 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns
 124 on two ingredients, and this paper takes up both for elasticity. The first is the
 125 polarisability. The Clausius–Mossotti value is the static response of a sphere;
 126 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that
 127 an infinite lattice propagates waves as the continuum does to a given order
 128 in kd ; Rahmani et al. [12] instead derive it from the exact static solution for
 129 the whole body, so that the method is exact in the long-wavelength limit.
 130 The second is the coupling. Integrating the Green tensor over the source
 131 cell rather than sampling it at the centre improves the accuracy, most of
 132 all for strong contrasts [13], and the filtered coupled dipoles of Piller and
 133 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace
 134 the point dipole by a source of finite extent whose spectrum is band-limited
 135 to the grid. The coupling summed over a lattice extends the method to
 136 periodic structures [16]. Its error converges at second order in the dipole
 137 spacing when shape errors are absent [3], and for a cube, the one shape
 138 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach
 139 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the

140 mathematical side, Costabel et al. [18] show that the method is a finite section
 141 of an infinite block Toeplitz matrix, stable for a large range of the contrast
 142 but not for all of it, and Costabel [19] that in a one-dimensional model it
 143 converges at second order in the interior, with a boundary layer in which
 144 the consistency error does not vanish. The layer studied here is the elastic
 145 counterpart of that cube: a body the grid represents exactly, so that the
 146 discretisation error can be isolated. What this paper adds is the closed form:
 147 for cells that tile space, the coupling that makes the lattice reproduce the
 148 continuum exactly, and the order and the constant of the error that remains,
 149 as functions of the two bases. In the terms used here the coupled-dipole
 150 method is the pair $p = r = 0$, and its refinements improve the polarisability
 151 and the coupling while leaving both bases constant in the cell.

152 *Its elastic counterparts.* The elastic problem has the same volume integral
 153 equation [20], and every cell-based solver of it is a coupled-dipole method
 154 whether or not it carries the name. Nestor [21] divided a two-and-a-half-
 155 dimensional medium into cells that interact through the background Green's
 156 tensor, each carrying a contrast renormalised by its own near field, with
 157 the multiple-scattering series summed. Kanaun and Levin [22] discretise the
 158 inclusion problem with Gaussian approximating functions whose matrix ele-
 159 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral
 160 equation on a grid by conjugate gradients with fast Fourier transforms, and
 161 Touhei [24] does the same for a half space with a generalised Fourier trans-
 162 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and
 163 inverse problems on the transition operator of the discretised Lippmann-
 164 Schwinger equation, the first in the acoustic approximation and the second

165 in arbitrary anisotropic elastic media with a nine-component state vector.

166 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and is
167 ordered as the experiment is built. §§2–3 set out the discrete medium and the
168 continuous layer it must reproduce; §4 shows why the point coupling fails,
169 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
170 single-site T -matrix as the first member of a Cartesian multipole hierarchy in
171 closed form, and sets out what that gives (§6.7). §§7–8 take up the wavefield
172 basis, §9 the medium basis and its projection error, §10 the sphere, and
173 §11 the hierarchy as a voxel scheme. §12 discusses what the results imply
174 and §13 concludes. Every result is derived symbolically in Mathematica and
175 checked against an independent Python/Fortran implementation or an exact
176 solution.

177 2. Setting

178 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
179 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
180 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
181 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
182 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
183 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

184 displacement followed by engineering strain. A voxel responds with a force
185 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
186 state to them, and the 9×9 lattice Green’s tensor $\mathbf{K}(i, j)$ maps a source

187 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

188 with ψ^0 the incident state.

189 *Where the kernel comes from.* Write the medium as the background plus
 190 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 191 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 192 ting $\rho = \rho_0 + \Delta \rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta \lambda \delta_{bc} \delta_{de} + \Delta \mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 193 the contrast stiffness, and moving the contrast terms to the right leaves the
 194 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

195 a force from the density contrast, and the divergence of the contrast stress m ,
 196 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the
 197 background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
 198 integrates by parts,

$$\int G_{ab}(\mathbf{x} - \mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

199 the boundary term vanishing because m is confined to the scatterers, and the
 200 last step holding because $\mathbf{G}$ depends only on the separation. The moment
 201 therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when
 202 the moment is the contrast stress itself. With the moment tensor defined
 203 instead through its equivalent body force $-\partial_c [M_{bc} \delta(\mathbf{x} - \mathbf{y})]$, as in Aki and
 204 Richards [27], the same field reads $M_{bc} \partial G_{ab} / \partial y_c$, the source derivative, with

205 $M = -m$. The two are one convention written two ways; what matters is that
 206 the derivative in the kernel and the sign in the contrast operator are chosen
 207 together. The nine-component kernel follows (Fig. 1): its force columns are
 208 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 209 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
 210 those entries and its strain rows their receiver derivatives, symmetrised in
 211 the same way and doubled on the shear rows to give engineering strain. In
 212 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

213 where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$
 214 and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

215 *Averaging, and collocation.* Three operations in this paper are averages over
 216 a cell, and they are distinct. The *source-cell average* integrates the kernel
 217 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 218 of a uniform polarisation of the source cell. Every scheme in this paper uses
 219 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 220 is imposed: at the receiving cell's centre, with the internal field of each cell
 221 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 222 *average* imposes the equation in the mean over the receiving cell instead;
 223 with a uniform internal field that is the Galerkin scheme of degree zero,
 224 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 225 term that refers to the internal field and not to the test, and they differ only
 226 in the test function (a delta at the centre against the cell's indicator) and
 227 in the constant of their $O((kd)^2)$ error (Table 8). The *cell-mean contrast*,

228 finally, is an average of the material, not of the field: where the medium
 229 varies within a cell, the contrast is projected onto the cell’s basis, its mean
 230 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 231 of the coupled-dipole method, the point dipole is collocation with the point
 232 kernel, the integrated Green tensor adds the source-cell average, and the
 233 filtered coupled dipoles replace the point source by one of finite extent.

234 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 235 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 236 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 237 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 238 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

239 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 240 join the centres of the voxels in one plane (a square lattice of pitch d), and
 241 m is the separation of the two planes in units of d (Fig. 2a,b). The question
 242 is whether, and how fast, this chain converges to the continuous layer as
 243 $n \rightarrow \infty$ at fixed D .

244 *Implementations.* Every result is derived symbolically in Mathematica and
 245 compared with an independent numerical implementation or an exact so-
 246 lution. “The package” below is the Python/Fortran implementation of the
 247 voxel scheme (`cubic_scattering`): its cube T -matrix, its lattice kernels and
 248 its stratified propagator, which serve both as the object under test and, where
 249 validated, as a second implementation.

250 3. The continuous layer and its thin-layer expansion

251 At normal incidence a plane force drives a one-dimensional problem per
 252 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 253 Green's function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer's reflection and transmission
 254 follow from continuity of displacement and traction at its faces; they agree
 255 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 256 P and S (`ContinuumLimit_Reference.wl`).

257 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 258 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 259 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 260 cient expands as

$$R = \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

261 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 262 coefficient symbolically and the truncated series against the exact layer).
 263 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

264 to first order a thin layer is an interface across which the traction jumps by
 265 the excess inertia and the displacement by the excess compliance. At oblique
 266 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 267 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 268 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients

269 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 270 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 271 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 272 a single plane of voxels must reproduce order by order. A plane of voxels
 273 that carry Legendre moments of their internal field to degree p reproduces
 274 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 275 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 276 first-moment voxel all four terms shown.

277 4. The specular sums of the point Green's tensor

278 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 279 of every voxel of a plane m planes away. In that form it is a poor object to
 280 study. The Green's tensor decays slowly with distance, so the sum converges
 281 slowly, and it hides the one question that matters here: which part of it is
 282 the continuum, and which part is the discretisation. The Poisson summation
 283 formula [28] answers both at once, and is the standard tool for lattice sums
 284 of wave fields [29]. For a function f on the plane and the square lattice of
 285 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

286 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 287 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).
 288 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

289 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 290 between planes a distance z apart. Each term has a physical meaning. Trans-
 291 formed laterally, the field of a plane of sources is a superposition of plane
 292 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 293 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 294 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 295 far beyond the wavenumber of the field when the voxels are small compared
 296 with the wavelength, so it is evanescent and decays away from the plane as
 297 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 298 the lattice sum exactly into the continuum and a discretisation correction,
 299 and the correction converges after a few orders.

300 *The local term that survives.* In the strain-from-moment block the evanescent
 301 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 302 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 303 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 304 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 305 planes in slab order, the static lattice sum of the point kernel is exactly
 306 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 307 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 308 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 309 reproduces the continuum only if that remainder is cancelled by the voxel's
 310 own self term.

311 5. The tiling identity

312 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 313 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 314 Each Poisson order in Eq. (10) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

315 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 316 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 317 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube’s indicator χ : a partition of unity (Fig. 4).
 318 Two exact statements follow.

319 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

320 the continuum’s plane-wave term times the vertical form factor. The pack-
 321 age’s cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 322 `AveragedSums.wl`).

323 *Within a plane.* The sum over *all* cells of the plane, the voxel’s own included,
 324 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

325 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 326 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 327 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 328 three normal strain-from-moment entries, whose local weights $-1/M_P$ and

329 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 330 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 331 with every cell integral evaluated analytically (below).

332 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 333 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

334 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 335 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 336 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 337 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 338 and the quadrature order are raised; Eq. (14) is now the package’s kernel at
 339 normal incidence.

340 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 341 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 342 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 343 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 344 which never contain the origin, and there the two-dimensional antiderivative
 345 in the face coordinates is continuous, so the face integral is a sum over the
 346 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 347 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 348 *not* usable for repeated derivatives: its arctangent terms jump across the
 349 coordinate planes through the origin, and differentiating twice there discards
 350 exactly that term.) The self term so obtained reproduces the classical closed
 351 form of the cube’s depolarisation [30, 31] to machine precision.

352 6. The single site: the cube T -matrix from moment integrals

353 Every error of a voxel scheme is made at the single site (§12), so the single
354 site is derived here rather than imported. The classical route imports it. For
355 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
356 uniform load uniform, with a depolarisation tensor in closed form [4, 32, 33];
357 its dynamic generalisation is known for spheres and ellipsoids [34, 35, 36].
358 For a cube or a general polyhedron the static field of a uniform eigenstrain
359 is also in closed form [30, 37, 31], and it has been computed for a cuboid in
360 an anisotropic matrix [38], but the strain it produces is not uniform, so there
361 is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors
362 for polyhedral inclusions with polynomial eigenstrains have since been given
363 [39], the dynamic inclusion problem has been formulated for shapes other
364 than the ellipsoid, cubes among them [40], and the Green’s tensor has been
365 integrated analytically over a cuboid for the coupled-dipole method, to an
366 error of $O((kd)^4)$ [41]; these are the static and electromagnetic counterparts of
367 individual moments below. In scattering theory the volume integral equation
368 of an elastic inclusion [20] is closed in the same spirit by a self-consistent
369 or effective-field approximation [42, 43]: a first-order (Born) scatterer [44]
370 renormalised by the geometric series of its own near field, the step that turns
371 a contrast into an effective contrast. Nestor [21] introduced the single site
372 in this form, summing the series of the cell’s self integral, with the singular
373 near field evaluated over an infinitesimal cell. The moments below evaluate
374 the same self integral over the finite cube, as a distribution. The elastic T -
375 matrix of Waterman [45] represents a scatterer in spherical multipoles, the
376 natural basis for a sphere; a voxel in a lattice needs one compatible with the

377 lattice, and the moments over the cube provide it. The construction below
 378 makes the approximation of a voxel explicit rather than implicit: it derives
 379 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 380 over the cube, with the uniform-strain closure as the first truncation of a
 381 hierarchy.

382 *A Cartesian multipole hierarchy.* We call the construction the Cartesian mul-
 383 tipole hierarchy (Fig. 5). The contrast acting on the voxel's field is a source:
 384 the density contrast a body force $\omega^2 \Delta \rho \mathbf{u}$, the stiffness contrast a stress
 385 $\delta \mathbf{c} : \boldsymbol{\varepsilon}$. Each voxel radiates through the Cartesian multipole moments of
 386 these sources over the cube, a force monopole and a stress dipole from the
 387 uniform field (the cell of the coupled-dipole method) and higher multipoles
 388 from the field's gradients, and receives the field through its gradients at its
 389 centre. The name needs one caution. A multipole expansion usually expands
 390 the kernel or the field about a centre and truncates it, which holds only for
 391 well-separated cells or outside a circumscribing sphere: in the fast multipole
 392 method [46], whose elastodynamic form leaves adjacent cells to direct inte-
 393 gration [47], in the spherical-wave T -matrix of Waterman [45], and in the
 394 force-dipole description of a point defect, an expansion of the Green's func-
 395 tion far from it [48]. Here the Green's tensor is never expanded. Only the
 396 source is resolved into Cartesian moments, and the integral over the source
 397 cube is exact at every separation, so there is no radius of convergence, and
 398 touching voxels and the self term, a distribution in closed form, are included.
 399 The source side is shared with the multipole extensions of the coupled-dipole
 400 method in optics [49, 50] and with Cartesian multipole decompositions of a
 401 source [51]. The word hierarchy has its sense in kinetic theory and cosmology

402 [52]: a chain of coupled moment equations closed by truncation at a chosen
 403 order, whose truncation error is here in closed form.

404 *The multipole orders.* A hierarchy of degree ℓ holds the force density $\omega^2 \Delta \rho \mathbf{u}$
 405 to degree ℓ and the stress $\delta \mathbf{c} : \boldsymbol{\varepsilon}$ to degree $\ell - 1$. A stress moment of degree n
 406 acts on the medium through its divergence, as a force multipole of order $n + 1$,
 407 so both channels carry the force multipoles through order ℓ (Table 1). Here
 408 a multipole is the primitive Cartesian moment, which keeps its traces, rather
 409 than the irreducible one [51]. What sets the order of convergence is each
 410 channel's first omitted source term, counted in its own degree (§6.4): fourth
 411 order needs the force quadrupole for a density contrast, but the second-degree
 412 stress moments, a force octupole, for a stiffness contrast.

Table 1: The force multipoles a hierarchy of degree ℓ carries, and the order of convergence it reaches for each contrast on the layer (Table 3).

ℓ	density channel	stiffness channel	order reached	
	(force moments)	(stress moments)	density	stiffness
1	monopole, dipole	dipole	2	2
2	to the quadrupole	dipole, quadrupole	4	2
3	to the octupole	to the octupole	4	4

413 *Its origin in the equivalent inclusion method.* In its static form the con-
 414 struction is the equivalent inclusion method with polynomial eigenstrains
 415 of Moschovidis and Mura [53]: the eigenstrain is expanded about the centre
 416 of the inhomogeneity, the consistency equation is expanded there in a Taylor
 417 series, and the coefficient of each monomial is equated, which amounts to

418 imposing the Lippmann–Schwinger equation and its derivatives at the cen-
 419 tre [54, §2.2]. Johnson et al. [55, 56] applied an expansion about the origin
 420 to a cuboidal inhomogeneity, and Wu et al. [39] expanded the eigenstrain
 421 of a polyhedron to second degree about its centroid; both report that the
 422 expansion about the centre does not follow the field near the edges, corners
 423 or vertices. Fu and Mura [34] extended the method to time-harmonic waves
 424 for an ellipsoid, with a second unknown for the density mismatch, and in the
 425 cases they report evaluated it with uniform eigenstrains. Brisard et al. [54]
 426 replaced the Taylor form by a Galerkin form of the same equation, because
 427 the Taylor form need not give an invertible system and a higher degree need
 428 not improve it. What is added here, to our knowledge, is the elastodynamic
 429 hierarchy, with density and stiffness contrasts together, to any degree and for
 430 a body that is not an ellipsoid; its moments as distributions in closed form;
 431 and the structure and the error of its truncations, tested degree by degree
 432 against the exact solution for a sphere. The moments here are integrals of
 433 the Green’s tensor against monomials over the body. They are not moments
 434 in the sense of the method of moments [57], of which the hierarchy is the
 435 instance with monomials about the centre as expansion functions and the δ
 436 function and its derivatives at the centre as testing functions; and the exact
 437 static response of an inclusion of any shape to polynomial loads is described
 438 by the polarisation and elastic moment tensors of Ammari and Kang [58].

439 *6.1. The closed hierarchy*

440 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 441 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. The hierarchy is stated below for any degree; its third degree
 442 is written out first, being the lowest at which the stiffness channel converges

443 at fourth order (§6.4). Expand the internal displacement about the centre to
 444 third degree and project the Lippmann–Schwinger equation onto 1, x_p , x_px_q
 445 and $x_px_qx_v$. The unknowns are $u_i(0)$, $\partial_p u_i(0)$, $\partial_p \partial_q u_i(0)$ and $\partial_p \partial_q \partial_v u_i(0)$, and
 446 the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$\begin{aligned} & (\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j \\ & + (\frac{1}{6} \omega^2 \Delta \rho \mathbf{B}_{ij,p}^{rst} - \frac{1}{2} \mathbf{P}_{in,pk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \end{aligned} \quad (16)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

$$\begin{aligned} & (\omega^2 \Delta \rho \mathbf{Q}_{ij,pqv}^r - \mathbf{X}_{in,pqvk} \delta c_{nkrj}) \partial_r u_j \\ & + (\delta_{pr} \delta_{qs} \delta_{vt} \delta_{ij} + \frac{1}{6} \omega^2 \Delta \rho \mathbf{J}_{ij,pqv}^{rst} - \frac{1}{2} \mathbf{Z}_{in,pqvk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \end{aligned} \quad (18)$$

447 all fields evaluated at the centre. Dropping the third gradients, that is
 448 Eq. (18) and the second line of Eq. (16), leaves the system of second de-
 449 gree; dropping the second gradients as well, that is Eq. (17) and the $\partial \partial u$
 450 terms of Eq. (15), leaves the uniform-strain closure. Equations (15) and (17)
 451 are the same at second and third degree, because every term the third de-
 452 gree would add to them vanishes by parity (§6.3). The coefficients are ten

453 moments of the background Green's tensor $\mathbf{G}$ over the voxel

$$\begin{aligned}
\mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\
\mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \\
\mathbf{B}_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{J}_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{X}_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} \, dV, \\
\mathbf{Z}_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s \, dV,
\end{aligned} \tag{19}$$

454 the first six being those of the second degree and the last four those the third
455 gradients add. The derivatives are taken with respect to the source point,
456 $\partial' = -\partial$. That sign reaches the moments with an odd number of derivatives,
457 $\mathbf{N}$, $\mathbf{Q}$, $\mathbf{B}$ and $\mathbf{J}$, and no symmetry, parity or scaling check can detect it, so it
458 is fixed by convention. The moments are graded by (D, W) , the number of
459 derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$, $\mathbf{M}(2, 0)$,
460 $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$, and $\mathbf{B}(1, 3)$, $\mathbf{J}(3, 3)$, $\mathbf{X}(4, 0)$, $\mathbf{Z}(4, 2)$.

461 *The hierarchy at any degree.* Nothing in the construction is special to the
462 third degree. Let the internal displacement be expanded to degree ℓ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \tag{20}$$

463 the sum running over ordered sets of indices $W = (w_1, \dots, w_m)$, and let the
464 strain be its derivative. Write the moments with any number of derivatives
465 and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \tag{21}$$

466 the derivatives being those of G_{in} as a function of its argument and the pairing
467 distributional (§6.2). The ten moments of Eq. (19) are these $\mathbf{g}$ themselves:

they are the integrals of $\partial'_D G_{in}(\mathbf{0} - \mathbf{x})$, and the factor $(-1)^{|D|}$ of the source derivative cancels the parity $(-1)^{|D|}$ of $\partial_D G_{in}$, so that, for example, $\mathbf{N}_{in,k}^r = \mathbf{g}_{in}[k; r]$ and $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$. Beyond the third degree the moments are written as $\mathbf{g}$ only. Imposing the Lippmann–Schwinger equation and its derivatives to order ℓ at the centre gives, for every set of derivative indices P with $|P| \leq \ell$,

$$U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \quad (22)$$

Equations (15)–(18) are the case $\ell = 3$, with the moments of Eq. (19) put for the $\mathbf{g}$; their truncations above are $\ell = 2$ and the uniform-strain closure $\ell = 1$. With the field rather than the eigenstrain as the unknown, the moments $\mathbf{g}[D; W]$ play the part of the polynomial Eshelby tensors of the equivalent inclusion method [33, Ch. 4], which Brisard et al. [54] call influence pseudotensors, here for the elastodynamic kernel and for the density channel as well. For an ellipsoid the static self term of a polynomial eigenstrain is itself a polynomial of the same degree, so the expansion about the centre is exact in the self term and approximate only in the interaction with other bodies [54]; for a cube it is approximate in both. The cube is centrosymmetric, so a moment vanishes unless $|D| + |W|$ is even, and the system needs the grades

$$\begin{aligned} (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\ (D, W) &= (m+1, W), \quad W \leq \ell-1 && \text{(stiffness),} \end{aligned} \quad m = 0, \dots, \ell, \quad D+W \text{ even.} \quad (23)$$

Table 2 lists what each degree adds. The unknowns number $3 \binom{\ell+3}{3}$: 12, 30, 60 for $\ell = 1, 2, 3$.

Table 2: The grades (D, W) of moments, D derivatives on the Green's tensor and a weight of degree W , that the hierarchy of degree ℓ needs beyond those of degree $\ell - 1$.

ℓ	new grades	moments in all
1	(0, 0), (1, 1), (2, 0)	3
2	(0, 2), (2, 2), (3, 1)	6: G, K, N, M, P, Q
3	(1, 3), (3, 3), (4, 0), (4, 2)	10: with B, J, X, Z
4	(0, 4), (2, 4), (4, 4), (5, 1), (5, 3)	15
5	(1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4)	21

Two independent systems. In either sum of Eq. (22) the unknown $U_{i;P}$ is coupled to $U_{j;W}$, or to $U_{j;rW}$, through a moment of grade $(|P|, |W|)$ or $(|P| + 1, |W|)$, and on a centrosymmetric body that moment vanishes unless $|P|$ and the degree of the unknown it multiplies have the same parity. The hierarchy of any degree therefore separates exactly, at every frequency and contrast, into two systems: one for the even gradients $u, \partial\partial u, \dots$ and one for the odd gradients $\partial u, \partial\partial\partial u, \dots$. Raising the degree from odd to even adds a block to the even system alone, and from even to odd a block to the odd system alone. At $\ell = 3$ the even system is Eqs. (15) and (17), unchanged from $\ell = 2$, and the odd system is Eqs. (16) and (18). Of the sixty unknowns, twenty-one belong to the even system and thirty-nine to the odd. The leading radiation of a density contrast, $\int_V \Delta \rho u \, dV$, is drawn from the even system, and that of a stiffness contrast, $\int_V \delta c : \nabla u \, dV$, from the odd one. The separation is thus the structure behind the result of §6.4, that the blocks improve the density and the stiffness channels alternately. At first degree it is implicit in the dynamic equivalent inclusion method for an ellipsoid, where the force and

503 the strain systems uncouple at the centre [34]; here it holds at every degree,
 504 for any centrosymmetric body.

505 6.2. The moments as distributions

506 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (24)$$

507 [59, 27]. Expanding g_c in powers of r reduces every moment in Eq. (19) to a
 508 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (25)$$

509 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 510 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 511 gral depends on the shape of the region excluded around the origin. This is
 512 the source of the δ -function term of the inclusion problem, which a numerical
 513 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 514 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 515 are distributional from the start, and each is moved onto the voxel's faces by
 516 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (26)$$

517 The surface term is evaluated on the faces, a distance h from the origin, where
 518 every derivative of r^m is smooth. The recursion lowers D and W together
 519 and ends on convergent integrals. The face integrals are done in closed form
 520 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (27)$$

521 which the face route satisfies exactly and a spherical excision returns as
 522 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
 523 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 524 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 525 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 526 Because every power of r is paired this way, the dynamic moments keep the
 527 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
 528 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 529 those terms, at a real $O((kh)^2)$ cost (§6.6).

530 *Every grade, in closed form on four constants.* Closed forms for monomial-
 531 weighted integrals of $1/r$ over bricks are known [60], as are closed forms for
 532 the potential and its first two derivatives over polyhedra with polynomial
 533 density, from the gravity literature [61]. The engine used here differs in
 534 taking derivatives of any order as distributions, in treating the kernel r as
 535 well as $1/r$, and in the even-power terms of the dynamic series. The recursion
 536 (26) does not depend on the grade, so the same engine gives every scalar
 537 moment of Table 2. Each is evaluated once and stored. The scalar moments
 538 of every grade through $\ell = 5$, with up to six derivatives and weights to
 539 the fifth degree, have been evaluated for the kernels r^{-1} and r : 4224 closed
 540 forms. (The second term of Eq. (24) places two further derivatives on r ;
 541 those moments are stored through $\ell = 3$ and are evaluated numerically, by
 542 the second route below, beyond it.) Every one of them is a power of the
 543 voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (28)$$

544 The integration returns many of them in other forms, with $\log(\sqrt{3} - 1)$,
545 $\log(1 + \sqrt{3})$, inverse hyperbolic tangents and logarithms of powers of two,
546 which are equal in value but not canonical: two equal moments can then differ
547 in appearance, and sums of large logarithms that cancel lose digits. Each
548 stored form is therefore reduced to Eq. (28) by an integer-relation search at
549 400 digits, with $\log 2$ and $\log 3$ admitted to the search, and the reduction
550 is accepted only if it agrees with the stored form to 1200 digits. All 4224
551 reduce, and the coefficients of $\log 2$ and $\log 3$ are zero in every one.

552 Every grade is gated by two checks whose answers do not depend on
553 the engine, the Laplacian sum rules of Eq. (A.1), which test the δ function
554 of the inclusion problem and its derivatives where a quadrature fails, and
555 confirmed by a second route that shares nothing with the faces of the cube,
556 to 1.1×10^{-12} or better (Appendix Appendix A).

557 6.3. Parity, and the nine-component block

558 The cube is centrosymmetric. A moment that couples a field quantity of
559 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
560 $\partial\partial u$ have degrees 0, 1 and 2, so at second degree Eq. (16) contains the first
561 gradient alone: ∂u is determined by its own block and couples to neither u
562 nor $\partial\partial u$. At any degree it couples to nothing of even degree; from $\ell = 3$
563 it couples to $\partial\partial\partial u$ (the second line of Eq. (16)). The leading scattering by
564 a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u .
565 The leading far field of the stiffness contrast is therefore fixed by Eq. (16),
566 and *no treatment of the second-gradient block (17) can change it*. What
567 Eq. (17) contributes reaches the far field only through the $O((kh)^2)$ coupling
568 in Eq. (15). That is the order, and the origin (the variation of the field across

the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes, as does the hierarchy itself once its third-gradient block is kept (§6.4).

With $\Delta\rho = 0$, Eq. (16) at second degree is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta C_{nkrj}$, built from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

The density channel is the displacement block of Eq. (15) with the second gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral of $\mathbf{G}$.

The cube against the sphere. For a sphere, Eshelby's tensor gives a dilatational channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's dilatational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equation makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on the shape. The cube's shear channels differ from each other. Their isotropic

586 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

587 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 588 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 589 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 590 and the diagonal shear is the more depolarised. At the background of §7
 591 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 592 0.4960.

593 *Where Eshelby enters.* Equations (29)–(31) follow from the moment integrals
 594 and the group theory alone. No ellipsoidal result is used, and no δ term is
 595 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 596 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 597 the uniformity theorem that ansatz is exact if and only if the inclusion is
 598 an ellipsoid. For a sphere the construction is therefore exact in the static
 599 limit, and for a cube it is an approximation. The hierarchy (22) states that
 600 approximation as the omitted blocks, rather than as an unquantified shape
 601 correction. A polynomial expansion about the centre cannot represent the
 602 edge and corner concentrations of an isolated cube's internal field at any finite
 603 degree; Table 5 measures what each degree leaves. A voxel in a uniform layer
 604 has no such concentrations: its neighbours have the same properties, so its
 605 internal field is the continuum's, which is smooth. That is why a low-degree
 606 internal field suffices in a lattice (§§7–8) although it is only an approximation
 607 for an isolated cube.

608 *Against the exact sphere.* These statements are tested against the exact
609 elastic Mie solution, with no Eshelby formula in between (Appendix Ap-
610 pendix F). Applied to a ball, the construction’s dilatational and shear chan-
611 nels equal Mie’s static concentrations to 2×10^{-10} , and the cube’s dilata-
612 tional channel and isotropic shear average meet the same numbers; taken
613 as a principal value the moment has no depolarisation at all, so the whole
614 static response is the δ term. At finite frequency the closure departs from
615 Mie only by a real term of order $(ka)^2$ at first order in the contrast, given
616 in closed form in Eq. (F.1); the radiation damping is exact. That term is
617 the uniform-field ansatz’s own error, the order the hierarchy assigns to the
618 omitted second gradients.

619 6.4. *What each block of the hierarchy buys*

620 A hierarchy invites the question of what its next block is worth. The
621 answer has a simple form, and it is not the one a count of unknowns suggests:
622 the blocks help in pairs.

623 *The experiment.* The layer of §7 at normal incidence is cut into n cells, and
624 in each the hierarchy of degree ℓ is imposed: the displacement is its Taylor
625 polynomial of degree ℓ about the centre, the strain the derivative of that, and
626 the equation and its first ℓ derivatives hold at the centre, with the δ terms
627 kept. In one dimension the moments are elementary and the exact layer is
628 known, so the order of each truncation can be measured for a density contrast,
629 a modulus contrast and both. The schemes are computed symbolically to 60
630 digits for $\ell = 1$ to 6 and, to $\ell = 3$, independently in double precision.

631 *The prediction.* A cell radiates the integral of its source against the kernel.
632 If the source is held to a polynomial of degree d about the centre, the first
633 power it omits is $d + 1$. When $d + 1$ is odd, that term integrates to zero
634 against the even part of the kernel and the error is two orders smaller; when
635 $d + 1$ is even it does not. The density source is the displacement, of degree ℓ ;
636 the stiffness source is the strain, of degree $\ell - 1$. So a density contrast gains
637 an order pair at every even ℓ , and a stiffness contrast at every odd ℓ . This is
638 the parity of §6.3, read as an order of convergence.

Table 3: The order of convergence of the hierarchy of degree ℓ on the layer, measured from 16 to 32 cells in reflection and in transmission at $\omega = 300 \text{ rad s}^{-1}$, and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

	$\ell = 1$	2	3	4	5	6
density contrast only	2	4	4	6	6	8
modulus contrast only	2	2	4	4	6	6
both	2	2	4	4	6	6
error, both, $n = 16$	5.4×10^{-6}	1.4×10^{-6}	5.8×10^{-12}	9.5×10^{-13}	2.6×10^{-18}	3.1×10^{-19}

639 *The result.* Table 3: all thirty-six orders are as predicted. The second gradi-
640 ents raise the density channel to fourth order and leave the stiffness channel
641 at second, with a constant three to four times smaller in reflection; the third
642 gradients bring the stiffness channel to fourth order; and so on in pairs.
643 With both contrasts, fourth order therefore needs $\ell = 3$ and sixth order
644 $\ell = 5$. The truncation of Eqs. (15)–(18) at second gradients is one block
645 short of the next order for a stiffness contrast, which is why it improves the

646 uniform-strain closure by a constant and not by an order.

647 *The error in closed form.* At first order in the contrast a cell whose source is
 648 held to its Taylor polynomial of degree d returns the exact term times $1 + E_d$,
 649 where, with h the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (33)$$

650 R and T standing for reflection and transmission, with $d = \ell$ for the den-
 651 sity term and $d = \ell - 1$ for the stiffness term. It follows from the series
 652 of $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$ with P_d the Taylor polynomial of e^{ikhx} , and it is
 653 confirmed symbolically for $d = 0$ to 6 and against the scheme on one cell,
 654 at a contrast of 10^{-20} and $kh = 0.03$, to 0.5% in all twelve cases. For
 655 $d = 0$ it is $(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission: the con-
 656 stants of the collocation scheme, Eq. (36), which are thus the first rung
 657 of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_`
 658 `layer_taylor_hierarchy.py`).

659 6.5. The single site against a sphere and a cube

660 The hierarchy is a statement about one scatterer, so it is tested first on
 661 one scatterer. The assembly of Eq. (22), with the full tensor moments in
 662 their dynamic form (the series of e^{ikr}/r taken to r^{11}), is the same for any
 663 centrosymmetric body; only the scalar moments change.

664 *A sphere.* For a ball the scalar moments are elementary, by the surface route
665 of Eq. (26) on $|\mathbf{r}| = a$, and the exact answer is known. A homogeneous
666 sphere of radius $a = 10$ m with the contrast of §7 is illuminated by a P plane
667 wave; the hierarchy’s polynomial source is radiated to nine scattering angles
668 between 11° and 169° , and the error is the largest component of the difference
669 from the exact displacement [62, 63] over the largest exact component. The
670 prediction, from the pairs of §6.4 with kh replaced by ka , is an error of
671 $O((ka)^2)$ for $\ell = 1$ and 2 and of $O((ka)^4)$ for $\ell = 3$, and $O((ka)^4)$ already
at $\ell = 2$ for a density contrast alone. Table 4 bears the prediction out. At

Table 4: One homogeneous sphere by the hierarchy of degree ℓ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

contrast	ℓ (unknowns)	$k_S a = 0.4$	0.2	0.1	order	predicted
density only	1 (12)	6.4×10^{-3}	1.6×10^{-3}	3.9×10^{-4}	2.0	2
	2 (30)	4.8×10^{-5}	2.6×10^{-6}	1.8×10^{-7}	3.9	4
	3 (60)	1.5×10^{-5}	9.5×10^{-7}	6.9×10^{-8}	3.8	4
moduli only	1	1.9×10^{-2}	5.0×10^{-3}	1.1×10^{-3}	2.3	2
	2	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	2.0	2
	3	6.2×10^{-5}	4.1×10^{-6}	2.0×10^{-7}	4.3	4
both	1	1.4×10^{-2}	3.1×10^{-3}	7.2×10^{-4}	2.1	2
	2	3.2×10^{-3}	6.9×10^{-4}	1.8×10^{-4}	1.9	2
	3	4.6×10^{-5}	2.5×10^{-6}	1.4×10^{-7}	4.2	4

672
673 $k_S a = 0.4$ the error of the uniform-strain closure is 1.4%, that of the second-
674 gradient system 0.32%, and that of the third-gradient system 0.0046%: three
675 hundred times below the closure. The systems are well conditioned, the sixty

676 equations of $\ell = 3$ having a condition number of 5.9 (`scripts/measure_`
677 `ball_gradient_hierarchy.py`).

678 *A cube.* The same assembly with the cube’s moments gives the single site of
679 a voxel. No exact solution exists for a cube, so the reference is the same cube,
680 of half-width $h = 10$ m, solved by the Legendre voxels of §8 on refined grids,
681 a cube being the one body such a grid represents exactly. Solutions with first
682 and second moments on 2^3 and 3^3 cells differ among themselves by at most
 2×10^{-5} , which is the accuracy of the reference. Table 5 shows what distin-

Table 5: One homogeneous cube of half-width h by the hierarchy of degree ℓ : relative far-field error against the converged reference.

ℓ (unknowns)	$k_S h = 0.4$	0.2	0.1	0.05
1 (12)	2.1×10^{-2}	5.4×10^{-3}	2.8×10^{-3}	2.7×10^{-3}
2 (30)	3.2×10^{-3}	1.6×10^{-3}	1.9×10^{-3}	2.4×10^{-3}
3 (60)	3.5×10^{-4}	3.4×10^{-4}	2.9×10^{-4}	3.1×10^{-4}

683
684 guishes a cube from a sphere. Each truncation has a floor that does not fall
685 with the frequency, because the internal field of an isolated cube is not a poly-
686 nomial even statically: it concentrates at the edges and corners, as the static
687 expansions about the centre of Johnson et al. [56] and Wu et al. [39] found for
688 the cuboid and for polyhedra. The floor is 2.7×10^{-3} for the uniform-strain
689 closure and nearly the same with second gradients, as §6.3 requires, the static
690 stress dipole being fixed by the first-gradient block and coupled next to the
691 third. The third-gradient block lowers it ninefold, to 3×10^{-4} . Above the
692 floor the error falls with the frequency as on the sphere: at $k_S h = 0.4$ the
693 third-gradient system is sixty times more accurate than the closure. The dip

694 of the second-gradient row at $k_S h = 0.2$ is a cancellation between its static
695 and dynamic terms, which have opposite signs. The condition numbers lie be-
696 tween 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These
697 results come from one implementation, checked against the exact sphere and
698 against the converged cube; the moments it uses are those confirmed by two
699 routes in §6.2.

700 6.6. The collocation closure

701 The collocation closure takes the internal field of a voxel as uniform, the
702 truncation of §6.3 for the density and stiffness channels together. The voxel's
703 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
704 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
705 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (34)$$

706 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)e$
707 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (35)$$

708 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
709 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
710 so Eq. (35) is precisely the collocation of the continuum Lippmann–Schwinger
711 equation with a strain piecewise constant over voxels. It does not contain
712 $\mathbf{S}_{\text{self}}$ at all.

713 The package's cube T -matrix is this closure statically: both blocks agree
714 with Eq. (34) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$

715 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about $0.25(kh)^2$
716 through its far-field form-factor corrections. Its self integrals are exact in the
717 static and radiation parts but omit the real even-power terms of the Green's
718 tensor's expansion ($k^2r, r^3, \dots$), an error of $0.4\text{--}0.6(kh)^2$ against the analytic
719 self term, whose series agrees with the closed-form elastodynamic tensor to
720 10^{-14} . By Eq. (35) that error cannot enter the closure scheme.

721 6.7. *What a closed-form single-site T -matrix gives*

722 The Cartesian multipole hierarchy costs more to set up than the alter-
723 natives: moments of the Green's tensor to fourth derivatives, evaluated as
724 distributions and reduced to closed form. What that buys follows from the
725 Lippmann–Schwinger equation of one body rather than from a model fitted
726 to it. *One derivation.* The static depolarisation, its dynamic corrections,
727 the radiation damping and every higher block come from that one equation
728 and one engine (§6.2), so they cannot be inconsistent with one another, and
729 the static, spherical and low-frequency limits are results of the construction
730 rather than conditions imposed on it. *Nothing is fitted.* Every coefficient
731 of Eq. (22) is a moment of the background Green's tensor over the body:
732 there is no polarisability to choose and no depolarisation factor to calibrate,
733 where other routes import the single site from the ellipsoid [4, 33] or close
734 the volume integral equation with an effective field [42, 43]. As in the equiva-
735 lent inclusion method [53, 34], the uniform-strain closure is the first member
736 (§6.3); what the hierarchy adds is the closed form at finite frequency for a
737 body that is not an ellipsoid, with the error of each truncation known. *The*
738 *singular self-interaction is exact:* the δ -function content that a quadrature
739 misses is part of the moments from the start, and the sum rule (27) and the

740 traces (A.1) hold at every grade (§6.2). *The exact limits are built in:* the
 741 cube's dilatational channel and its isotropic shear average equal the sphere's,
 742 confirmed against the exact Mie solution to 2×10^{-10} and 10^{-8} (§6.3), and
 743 at finite frequency the closure departs from Mie only through a real term
 744 of order $(ka)^2$, the radiation damping agreeing exactly in all three channels
 745 (Eq. (F.1)). *The error is stated before anything is solved:* a truncation is
 746 a statement of which blocks are omitted; the orders come in the predicted
 747 parity pairs, all thirty-six on the layer and all nine on the sphere (Tables 3
 748 and 4), and the leading error factor is in closed form, Eq. (33). *It improves*
 749 *block by block, on fixed constants:* raising the degree adds moments, con-
 750 firmed by a second route that shares nothing with the first (§6.2), and no
 751 quadrature; against the exact sphere at $k_S a = 0.4$ the error falls from 1.4%
 752 for the uniform-strain closure to 0.0046% with third gradients, from a sys-
 753 tem of sixty equations with a condition number of 5.9 (Table 4). That each
 754 degree improves on the one below is shown here for one sphere and one cube,
 755 not in general: in the Taylor form of the equivalent inclusion method, for
 756 interacting inhomogeneities, a higher degree need not improve the solution
 757 and the system need not be invertible [64, 54]. *It says what a body's shape*
 758 *does:* at the order of the uniform-strain closure a cube is a sphere plus a pure
 759 cubic anisotropy, which splits the deviatoric channel about the sphere's value
 760 in the ratio 3 : 2 (Eq. (32)). *It describes a scatterer, not only a cell:* the as-
 761 sembly is the same for every centrosymmetric body, only the scalar moments
 762 changing (§6.5), and the same moments are the self block of a voxel scheme
 763 (§11).

764 *What it does not give.* A polynomial about the centre cannot follow the
765 edge and corner concentrations of an isolated cube's internal field, so each
766 truncation there has a floor that does not fall with the frequency (3×10^{-4}
767 with third gradients; Table 5). And in a lattice the Legendre voxel of §8 is the
768 more economical in unknowns: both are fourth order on a smoothly graded
769 sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel
770 with thirty-six (§11). The hierarchy's case is not the number of unknowns;
771 it is that it yields the single site itself, in closed form, with its error known.

772 **7. Convergence, and the closed-form error**

773 *The experiment.* Every convergence test of this paper is the same experiment,
774 varied in one respect at a time, and it is set out here once.

- 775 1. *The medium.* The background has $\alpha = 5000$ and $\beta = 3000 \text{ m s}^{-1}$ and
776 $\rho = 2500 \text{ kg m}^{-3}$. The layer occupies $0 < z < D$ with $D = 2 \text{ m}$,
777 and differs from the background by $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$ and
778 $\Delta\rho = 100 \text{ kg m}^{-3}$: 6.4% in the P modulus, 4.4% in the shear modulus
779 and 4% in the density.
- 780 2. *The source.* A plane force at $z_s = -12 \text{ m}$, above the layer, radiates a
781 plane wave towards it. The incident field in the layer is then known in
782 closed form, and it is smooth there because the source is outside.
- 783 3. *What is measured.* The scattered displacement, total less incident, at
784 two observers outside the layer: at $z = -1 \text{ m}$ above it, where it is the
785 reflected wave, and at $z = 4 \text{ m}$ below it, where it is the change the layer
786 makes to the transmitted wave. Outside the layer the scattered field is

787 a single outgoing wave, so one observer on each side holds everything;
 788 the two are reported separately, as reflection / transmission.

789 4. *The exact answer.* The reflection and transmission of the layer from
 790 continuity of displacement and traction at its two faces (§3); for a
 791 layer whose properties vary with depth, from the differential equation
 792 integrated to 30 digits.

793 5. *The error.* $e = |u_{\text{scheme}} - u_{\text{exact}}|/|u_{\text{exact}}|$ at each observer: relative to the
 794 scattered field, not to the total, so that an error of 10^{-6} is a millionth
 795 of what the layer scatters.

796 6. *The refinement.* The layer is cut into n planes of voxels of side $d =$
 797 D/n , with n from 1 to 32, at $\omega = 60, 300$ and 600 rad s^{-1} , for which
 798 $k_P D = 0.024, 0.12$ and 0.24 and $k_S D = 0.04, 0.2$ and 0.4 . Three
 799 frequencies show that an order is not an accident of one.

800 7. *The arithmetic.* The errors of the higher-degree voxels fall below the
 801 round-off of double precision, so the symbolic implementation computes
 802 them to 40 digits; the independent numerical implementation, in double
 803 precision, agrees with it down to its own floor of about 10^{-13} .

804 The later sections change one item each: the incident wave to SV or SH and
 805 its angle (§8), the medium to a stratified or a graded one and the background
 806 to a stratified one (§9), and the body to a sphere (§10).

807 *Second order.* For a layer $D = 2 \text{ m}$ with the contrast $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu =$
 808 1 GPa , $\Delta\rho = 100 \text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000 \text{ m s}^{-1}$, $\rho =$
 809 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 810 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
 811 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (35) and with the

812 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
813 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
814 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
815 the exact solution and n the refinement; the orders in the legend of Fig. 7
816 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
817 with their standard errors. The two agree where a sequence is asymptotic
818 and differ where it is not, which the apparent orders show.

819 *The leading error in closed form.* In Eq. (35) the kernel is integrated exactly;
820 the only approximation is the internal field's variation across a voxel. At first
821 order in the contrast the discrete scattered field is therefore a midpoint rule
822 applied to the continuum's Born integral, with the kernel's cell integral exact.
823 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
824 it is flat. With k the background wavenumber of the wave concerned (k_P for
825 an incident P wave, k_S for an incident S wave), R the reflection coefficient
826 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (36)$$

827 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
828 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
829 and those of Eq. (39) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
830 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
831 the transmission constant grows to -0.092 .

832 *Why a uniform-field single site stops at second order.* The two constants have
833 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can

cancel both. The package T -matrix illustrates it: its form-factor corrections reduce the reflection error by a factor 4.45 and *increase* the transmission error by 1.7, at every n . The missing term in Eq. (36) is the first moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) = P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (37)$$

with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (38)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form. Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_`

FourthOrder.w1; Table 6). A single plane of first-moment voxels represents

Table 6: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (38). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

853

854 the 2 m layer 6500 times more accurately than a single plane of uniform ones.

855 The collocation column is the scheme of §6 and reproduces the chain of the

856 full 9×9 system to all printed digits.

857 *Higher moments.* Carrying the Legendre moments to degree p changes noth-

858 ing in the construction: the moments and the same-cell double integrals of

859 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-

860 degree solver reproduces the chain above exactly. The second-moment voxel

861 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at

862 eighth, in reflection and transmission (Table 7; `ContinuumLimit_SecondMoment.`

863 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .

864 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

865 *The leading error in closed form.* Without scattering, the Galerkin solution

866 is the L^2 projection Π of the incident field onto each voxel's basis, and the

867 kernel is integrated over each voxel exactly; at first order in the contrast the

Table 7: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} , $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (39)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 8. The chain of Eq. (38) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6} . The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 8: The leading error of the discrete layer in closed form, Eq. (39): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

882 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (40)$$

883 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
884 mission for odd p , opposite signs for even p . The first-moment voxel's leading
885 error therefore does not depend on the direction in which the field is observed.
886 Eq. (40) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
887 ishes) in Mathematica and, independently, in sympy.

888 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
889 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
890 $d = D$) the series in D of the reflection coefficient, computed exactly at the
891 full contrast, coincides with that of the exact layer, Eq. (7), term by term
892 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
893 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

894 `w1`). Each further Legendre moment adds two terms of the thin-layer series.
 895 *In three dimensions.* The three-dimensional first-moment voxel carries the
 896 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 897 in all, tested with the same functions and coupled by cell-to-cell double in-
 898 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 899 cidence it reduces *exactly* to Eq. (38) (`ContinuumLimit_Reduction3D.w1`).
 900 The lateral first moments are never excited: the lattice and the incident
 901 field are symmetric under reflection about every voxel's centre planes, and
 902 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 903 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 904 ishes on the reciprocal lattice and the plane sum of the double integrals
 905 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 906 three-dimensional structure begins. The lateral first moment's form fac-
 907 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 908 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 909 evanescent orders. And at oblique incidence the mean's own zeros shift,
 910 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 911 only to $O(k_{\parallel} h)$.
 912 *Other angles, other waves, and degree two.* The orders and their constants
 913 hold beyond normal incidence and incident P. At oblique incidence, where
 914 the tiling identity holds only to $O(k_{\parallel} h)$ and the gradient channels couple
 915 through the evanescent orders, the mean-only and first-moment voxels remain
 916 second and fourth order in the reflected P and the converted S wave; for
 917 incident SV and SH waves the orders, and with $k = k_S$ the constants, are
 918 those of incident P; and the voxel of degree two converges at sixth order

919 for P, SV and SH waves, its error on four planes at normal incidence being
920 the $c_2(kd)^6$ of Eq. (40) to 4%. Appendix Appendix B gives the spectral
921 construction, Eq. (B.1), the tests, and the second implementations that check
922 them; Figure 7(a,b) shows the convergence.

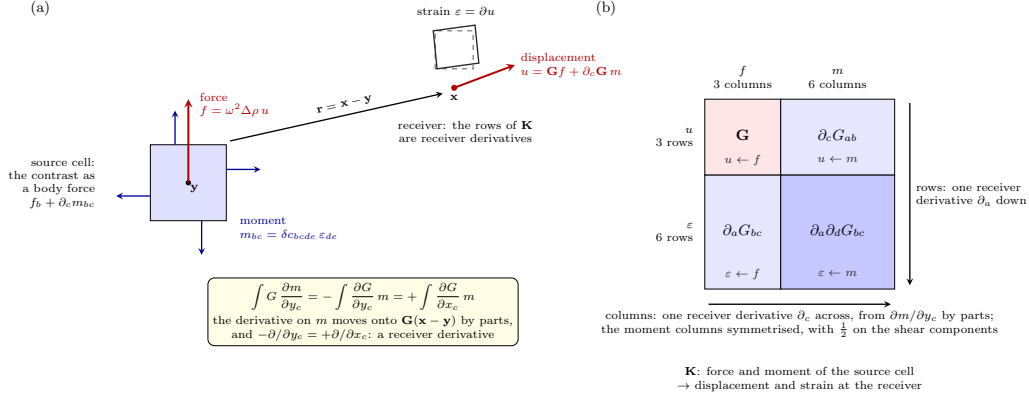


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

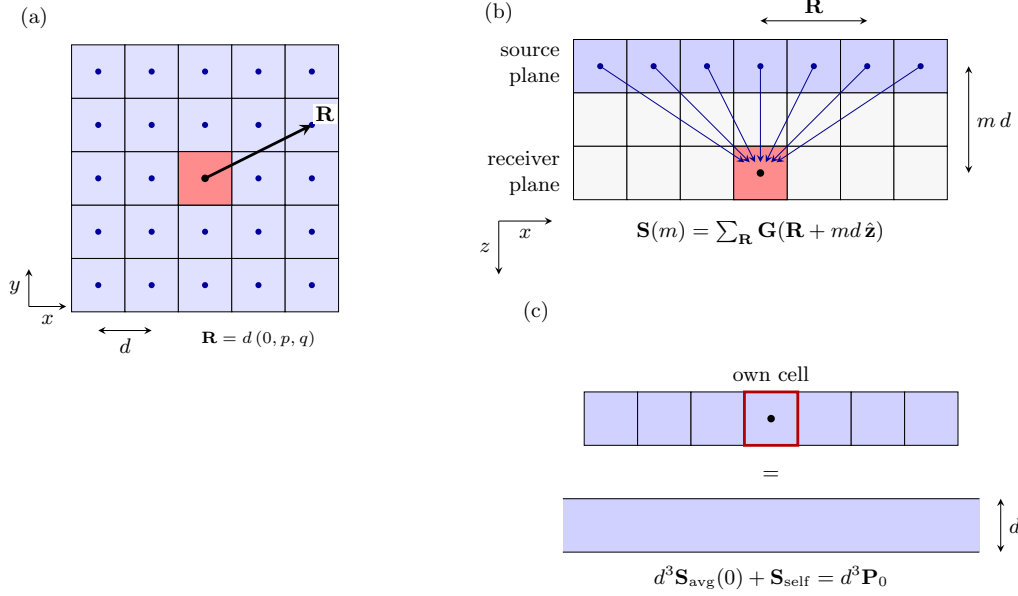


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

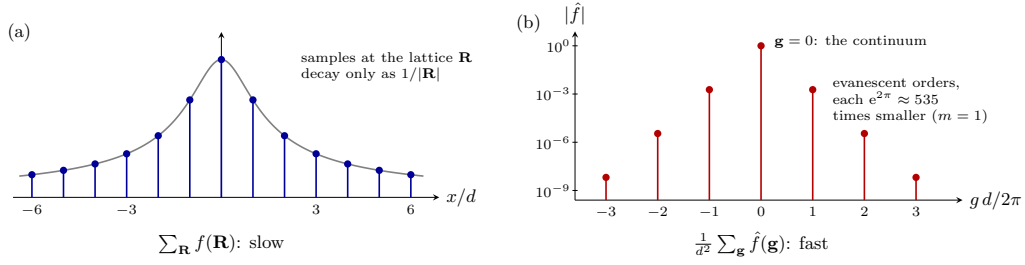


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

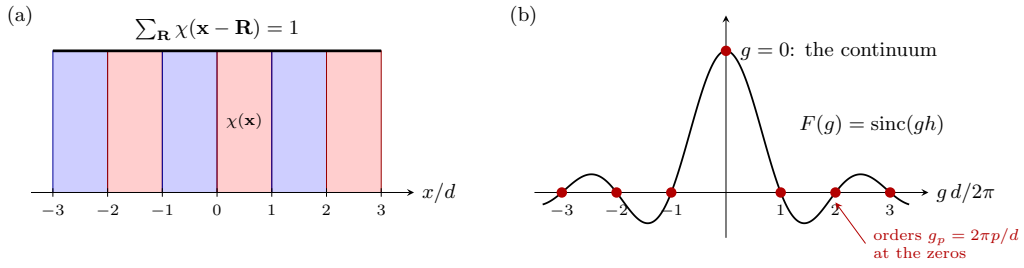


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

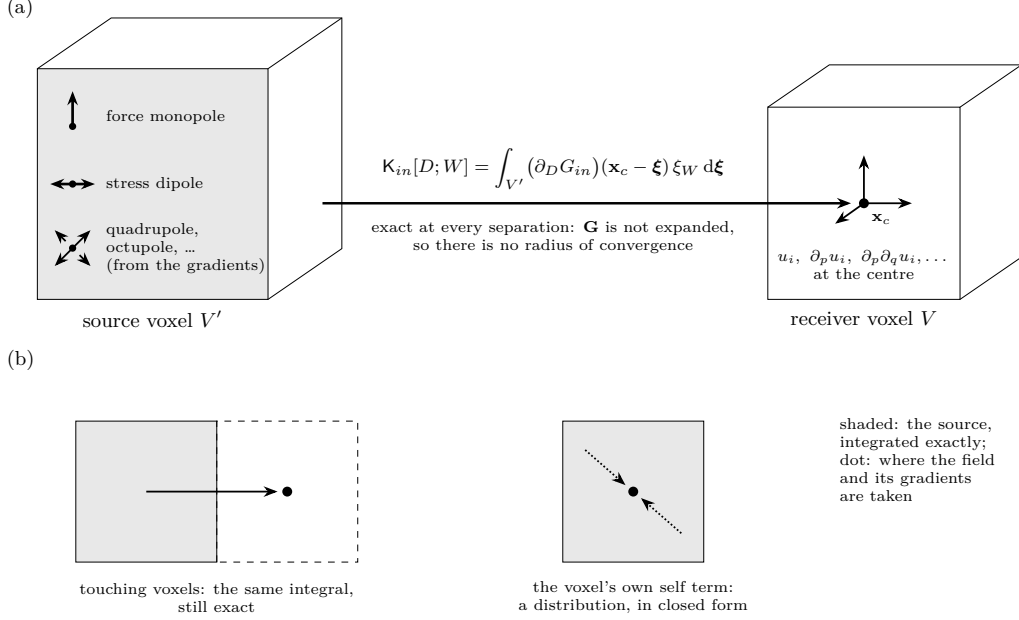


Figure 5: The Cartesian multipole hierarchy. (a) The source voxel radiates through the Cartesian multipole moments of its source: a force monopole from the density contrast, a stress dipole from the stiffness contrast, and the quadrupoles, octupoles and higher multipoles from the field's gradients. The receiver voxel takes the field through its value and gradients at its centre $\mathbf{x}_c$. The two are coupled by the integral of the derivatives of the Green's tensor over the whole source cube, Eq. (47), which is exact at every separation because the Green's tensor is not expanded. (b) The cases a multipole expansion of the Green's tensor cannot reach are therefore included: touching voxels, by the same integral, and the voxel's own self term, a distribution in closed form.

Alt text: Two panels. In the upper panel a shaded cube on the left, labelled source voxel, contains three small symbols: a single arrow labelled force monopole, a pair of opposed arrows labelled stress dipole, and a cross of four arrows labelled quadrupole, octupole, and so on, from the gradients. A long arrow runs from it to the centre of an unshaded cube on the right, labelled receiver voxel, where a dot with three short arrows marks the

field and its gradients at the centre. Above the arrow is the coupling integral of the derivatives of the Green's tensor over the source cube, and below it the words exact at every separation: $\mathbf{G}$ is not expanded, so there is no radius of convergence.

In the lower panel, on the left, a shaded square touches a dashed square whose centre is marked by a dot, with an arrow from the shaded square to the dot, labelled touching voxels: the same integral, still exact; on the right a single shaded square has a dot at its centre and two

923 9. A heterogeneous layer

924 A heterogeneous medium is not a layer of identical cells. The simplest
925 heterogeneous case with an exact answer is a stratified model built from
926 layers of constant properties, with real interfaces between them: the form of
927 many layered crustal velocity models. The last test takes such a model. A
928 layer $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each
929 one $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background
930 (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is
931 a random stratification with physical contrasts. The voxels refine the model's
932 layering, with m planes of voxels in each model layer, so no voxel straddles
933 a model interface: the natural discretisation when the model's interfaces are
934 known. At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so
935 the field reverberates between the model interfaces. The schemes are those
936 of §§7–8, with the contrast now changing from plane to plane; nothing else
937 in them changes. In the terms of the experiment of §7, the test is this.

- 938 1. *The medium.* The eight sets of contrasts are drawn once, from a fixed
939 seed, and the same eight are used in every run, so that every grid and
940 every scheme sees the same model.
- 941 2. *Normal incidence.* The source is the plane force at $z_s = -12$ m, and the
942 scattered displacement is taken at $z = -1$ m, where it is the reflected
943 wave, and at $z = D + 2 = 6$ m, where it is the change in the transmitted
944 wave. The P problem has the modulus contrast $\Delta\lambda + 2\Delta\mu$ and the wave
945 speed α ; the S problem is the same one-dimensional problem with $\Delta\mu$
946 and β .
- 947 3. *Oblique incidence.* A P plane wave is incident at 20° to the vertical.

948 What is measured is the pair of displacement amplitude vectors of
 949 the reflected P and the reflected S plane waves, R_{PP} and R_{PS} , in the
 950 specular order. The lateral sums of the coupling (§8) run over the
 951 reciprocal-lattice orders $|p|, |q| \leq 8$.

952 4. *The exact answer.* The state of displacement and traction, two compo-
 953 nents at normal incidence and four for P–SV at 20° , is carried across
 954 each model layer by the matrix exponential of that layer’s system ma-
 955 trix times its thickness, the eight matrices are multiplied in order,
 956 and the radiation conditions of the two half-spaces are imposed on
 957 the product. The arithmetic carries 30 to 40 digits. The reference
 958 shares nothing with the voxel scheme but the model. It is checked
 959 against a third algorithm, the reflection–transmission recursion of Ken-
 960 nett [65] over the nine interfaces: the same-type coefficients, R_{PP} and
 961 R_{SS} at normal incidence and R_{PP} at 20° , agree to 10^{-14} (`scripts/`
 962 `gate_heterogeneous_reference_vs_kennett.py`). Only same-type
 963 coefficients are compared, because they do not depend on how the up-
 964 and downgoing waves are normalised; a converted coefficient does. One
 965 sign differs, by convention: at normal incidence the one-dimensional
 966 model’s R_{PP} is a ratio of vertical displacements, the recursion’s a ra-
 967 tio of amplitudes along each wave’s own direction, and the reflected P
 968 wave travels upwards.

969 5. *The error.* As in §7, the difference from the exact scattered quan-
 970 tity relative to its size: for the amplitude vectors at 20° , $|\mathbf{R}_{\text{scheme}} -$
 971 $\mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$.

972 6. *The refinement.* Each model layer is cut into m planes of voxels, so

973 that the layer holds $n = 8m$: $m = 1, 2, 4, 8, 16$ at normal incidence
974 and $m = 1, 2, 4$ at 20° , where the lateral sums make each run dearer.
975 The orders are apparent orders, from $m = 4$ to 16 and from $m = 2$ to
976 4.

The errors are in Table 9 and Fig. 7(c).

Table 9: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

977

978 *The interfaces cost nothing.* Every order is the one measured for the uniform
979 layer. At a model interface the strain of the true field jumps. A voxel basis
980 that is discontinuous from voxel to voxel represents that jump exactly when
981 the interface lies on a voxel boundary, so the interface adds no error of its
982 own. The only requirement is that voxel boundaries coincide with the model's
983 interfaces. A voxel that straddled an interface would carry a contrast that is
984 not uniform within it, a case not examined here.

985 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
986 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
987 layer. That figure is not a run: it is the second-order law carried from the
988 finest grid computed, 128 planes, whose error is 7.3×10^{-6} , to the target,
989 $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$. The first-moment voxel needs 8, one per model
990 layer, and that figure is a run: the model's own layering already gives $8.5 \times$
991 10^{-7} . Counting unknowns (9 per voxel against 36), that is 3123 against 288
992 per vertical column of voxels, a factor of eleven. In three dimensions, with
993 the lateral grid refined as the vertical one is, the number of voxels falls by
994 $(347/8)^3 \approx 8 \times 10^4$ and the number of unknowns by about 2×10^4 . The same
995 model has been solved independently in Python, with a contrast for each
996 plane: the raw reflected displacements agree with the Mathematica scheme
997 to 1.1×10^{-15} at normal incidence and at 20° , with and without the first
998 moments. Lateral heterogeneity needs the full three-dimensional solver; it is
999 the first extension discussed below.

1000 *A medium that varies within a cell.* The stratified model is exact on its own
1001 grid: every cell is homogeneous. A smoothly varying medium is not, and
1002 sampling it as cells of constant properties is an approximation of its own.
1003 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
1004 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
1005 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
1006 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (38)

1007 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (41)$$

1008 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 1009 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 1010 the order of convergence is

$$\min(2p+2, 2r+2), \quad (42)$$

1011 or $2p+2$ when degree r represents the profile exactly (Table 10; `ContinuumLimit_`
 1012 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 1013 caps every voxel at second order, however rich its field basis, and for the
 1014 smooth profile the voxel of degree 2 with a linear contrast has the error
 1015 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 1016 this is what one expects: the contrast's projection error is orthogonal to
 1017 the polynomials of degree r , so against the smooth product of the inci-
 1018 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 1019 the contrast must be represented to the degree of the field. The closed forms
 1020 of the construction are tested one by one by an independent implementa-
 1021 tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
 1022 of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
 1023 linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
 1024 double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
 1025 bles the scheme without linearisation, evaluating the projected contrast at
 1026 the quadrature nodes, against its own 30-digit solution of the graded layer;
 1027 its errors agree with those of Table 10 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it

Table 10: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (42), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

1028 measures the orders of Eq. (42) again.

1029 *Which basis sets the error.* Equation (42) gives the order. The size of the
1030 error, and which of the two bases it belongs to, appear when the scattered
1031 field is separated by its order in the contrast: the Born term T_1 , and the
1032 second-order term T_2 in which the field scattered once is scattered again.

1033 Appendix Appendix G gives both terms for the exact layer in closed
1034 form (Eqs. (G.2)–(G.7)) and for the scheme, where the first needs no solve
1035 (Eq. (G.8)), and shows how both are read from solutions at four scaled
1036 contrasts (Eq. (G.9)). The terms must be separated: at the full contrast the
1037 terms of third and higher order are 7% of the second and carry errors of their
1038 own.

1039 With $f(z)$ the profile and Π_q the L^2 projection onto the Legendre poly-
1040 nomials of degree at most q in each cell, define the *relative projection error*

1041 of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (43)$$

1042 the fraction of the squared norm of the profile that polynomials of degree q
 1043 on the cells cannot represent. It is a property of the medium and the grid
 1044 alone; it involves neither the wave nor the solution. For the smooth profile
 1045 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (44)$$

1046 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 1047 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 1048 The error of the Born term is smaller by two to six orders of magnitude. The
 1049 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 1050 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 10
 1051 has 1.4×10^{-7} .

1052 Equation (44) is exact in the long-wave limit (Appendix Appendix G).
 1053 There the strain scattered once is the profile itself, a cell holds it as $\Pi_{\min(p,r)} f$,
 1054 and the second scattering takes place in the cell of the first: the law is
 1055 a statement about the single site, whose T -matrix is exact for the cell as
 1056 represented and whose error is that the cell is not the medium. Measured,
 1057 the part of T_2 between different cells carries at most 0.002 of the error, and
 1058 the departure from the law falls as $(kD)^2$.

1059 Three consequences follow. The two bases must be matched: a contrast
 1060 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 1061 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 1062 the contrast improves only the Born term. The wavefield basis is there to

1063 carry what the medium imprints on the field, not only the wave: at these
 1064 frequencies the wave itself varies little across a cell. And the error is known
 1065 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
 1066 and the cells. It vanishes when the medium is piecewise constant on the cells,
 1067 which is the stratified model above; there the contrast needs no more than a
 1068 constant in each cell, and the degree of the field alone sets the order.

1069 *A stratified background.* Everything above takes the background to be a
 1070 whole space. The method is meant for a background that is plane strati-
 1071 fied, with the voxels carrying the departure from it, and the results must
 1072 hold there. They do, in the one case tested: normal incidence, P, with the
 1073 background two half-spaces joined at $z = z_I$ below the layer, the lower one
 1074 faster and denser (displacement reflection coefficient -0.19). The voxels then
 1075 interact through the Green's function of that background,

$$g(z, z') = \frac{i}{2M_P k} \left[e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (45)$$

1076 with ρ_I the reflection coefficient of the interface: the whole-space term and
 1077 a reverberation that is smooth and carries no local term. The incident field
 1078 is the field of the source in the background, and the exact answer is the
 1079 reflection of the whole stack less that of the background. With the interface
 1080 3 m below the layer, and again with the lowest voxel against it, the scheme
 1081 converges at order 2.00, 4.00 and 6.0 for $p = 0, 1$ and 2 on the uniform
 1082 layer, and at 1.98, 3.98 and 5.98 on the smooth profile with $r = p$. The
 1083 first-order term is predicted to round-off (5×10^{-16}): it is the sum over
 1084 four paths, the incident wave travelling down or up and the scattered wave
 1085 leaving up or down, of the exact first-order term of each path times the

1086 factor $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$ for that path's pair
 1087 of wavenumbers, whose leading terms are the constants of Eq. (40). And the
 1088 relative error of T_2 over $\mathcal{E}_p$ is 1.00 to 1.02 at $kD = 0.12$ on 4 to 16 cells for all
 1089 three degrees, departing from one by 1.1×10^{-4} and 3×10^{-5} at $kD = 0.012$
 1090 for the two positions of the interface: Eq. (44) holds as in the whole space
 1091 (`scripts/measure_layer_stratified_background.py`; the exact stack is
 1092 computed in 30-digit arithmetic). The reason is the one given for Eq. (G.10):
 1093 the reverberation adds no local term, so the second scattering in the cell of
 1094 the first is unchanged. This check is by one implementation against an exact
 1095 solution.

1096 *Oblique incidence, and a three-layer background.* The same holds at oblique
 1097 incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer
 1098 background in which the layer of voxels reverberates between two interfaces
 1099 (Appendix Appendix C). At 20° the orders are 2, 4 and 6 for $p = 0, 1$ and
 1100 2 in all thirty-six cases (Table C.16), and the errors are close to those of the
 1101 whole space: the background changes the field the voxels must represent,
 1102 not how well they represent it. Both terms of the error remain known in
 1103 advance. The scheme's first-order term equals a closed-form prediction, built
 1104 from the thirty-six pairs of incident and radiated plane waves (Eq. (C.3)), to
 1105 2×10^{-15} ; and the law of Eq. (44) holds to 3% for $p \leq 1$ and 6% for $p = 2$ at
 1106 $k_S D = 0.2$, and to 0.6% at $k_S D = 0.067$, in every background and for every
 1107 wave type (Table C.17).

1108 10. A sphere: what the layer predicts in three dimensions

1109 The layer and the stratified layer are laterally uniform. A sphere is not,
 1110 and elastic scattering by a sphere, homogeneous [62, 63] or radially varying
 1111 [66], has an exact solution: it is the usual benchmark for volume-integral and
 1112 discrete-dipole methods. This section solves two spheres with the full three-
 1113 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
 1114 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
 1115 iterative solver, and a P wave is incident. For the graded sphere the voxel is
 1116 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
 1117 nine-component state, tested with the same constant, and the L^2 projection
 1118 of its contrast, which is its cell mean; every cell that overlaps the sphere is
 1119 kept, and the cells are coupled by cell-to-cell double integrals of the Green’s
 1120 tensor, one block per orbit of the cube group (Appendix Appendix E). For
 1121 the sharp sphere the voxel is the package’s cube T -matrix, which differs from
 1122 the collocation closure only through its far-field corrections (§6.6) and has
 1123 the same order, each cell keeping the contrast of its centre. The far field is
 1124 computed from the solution for the incident plane wave itself, each voxel’s
 1125 displacement and strain carrying the plane-wave phase at its centre; adding
 1126 to that phase a linear variation about the scatterer’s centre, as a Taylor
 1127 expansion of the incident field would, counts the wave’s variation twice and
 1128 leaves an error of order $(ka)^2$ that no refinement removes. In the terms of
 1129 the experiment of §7, the test is this.

- 1130 1. *The body.* A sphere of radius $a = 10$ m (Fig. 6) in the background of
 1131 §7, carrying the contrast of that section, at $\omega = 150$ and 300 rad s^{-1} ,
 1132 for which $k_S a = 0.5$ and 1 and $k_P a = 0.3$ and 0.6 .

- 1133 2. *The incident wave.* A P plane wave travelling along $+z$.
- 1134 3. *The grid.* The cube of side $2a$ that circumscribes the sphere is cut into
1135 n_{sub}^3 cells of side $2a/n_{\text{sub}}$; which of them are kept is stated with each
1136 voxel above.
- 1137 4. *The solve.* The iteration is carried to a relative residual of 10^{-11} , far
1138 below every discretisation error reported, so that the errors are the
1139 scheme's and not the solver's.
- 1140 5. *The exact answer.* The partial-wave series of the sphere, summed to
1141 order 10 at $k_S a = 0.5$ and 11 at $k_S a = 1$, and evaluated as the exact
1142 scattered displacement at the observers, not as its asymptotic form.
- 1143 6. *What is measured.* The scattered displacement vector at nine scattering
1144 angles, equally spaced from 11° to 169° in the z - x plane: forward and
1145 backward scattering and everything between, P and S together.
- 1146 7. *The error.* The largest component of the difference between the scheme's
1147 far field and the exact displacement over the nine angles, divided by the
1148 largest component of the exact one. It is a maximum, not an average,
1149 so that no angle is hidden.
- 1150 8. *The distance.* The solution of the scheme is the field inside the voxels,
1151 and the scattered field outside follows from it at any distance. For the
1152 far-field test it is read with the asymptotic formula, which keeps the
1153 term in $1/r$ alone: a choice of this test, not a property of the scheme.
1154 The reference is the exact field, so the two differ by the near-field terms,
1155 of relative size $1/(kr)$, however good the scheme. The graded sphere
1156 is therefore observed at 5×10^8 radii, where those terms are 10^{-8} of
1157 the peak: at 5×10^4 radii they are 1.4×10^{-4} , which is 4% of the

1158 discretisation error at 12 voxels across and most of it at 24, and would
 1159 be read as a failure to converge. The sharp sphere, whose errors are
 1160 a hundred times larger, is observed at 5×10^4 radii. The near field is
 1161 tested separately, with a readout that drops nothing (Table D.18).

1162 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 1163 its shape with every grid. A sphere whose contrast falls smoothly to zero at
 1164 its surface does not: the voxels that cross the surface carry a vanishing con-
 1165 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 1166 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 1167 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 1168 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 1169 the radial equations of Takeuchi and Saito [66], integrated across the shell
 1170 from the core’s regular solutions and matched to outgoing waves at $r = a$.
 1171 The equations are derived symbolically from the equations of motion, and
 1172 the solution is checked against a sphere of homogeneous concentric shells
 1173 in closed form, by unitarity and reciprocity, and by an independent imple-
 1174 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 1175 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 1176 carries the mean of the contrast over its cell. Table D.19 shows the error
 1177 falling at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/`
 1178 `pilot_graded_voxel_sphere.py`): the apparent order between successive
 1179 grids is 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares
 1180 orders of Fig. 7(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across
 1181 the error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 1182 graded shell and the order is not yet asymptotic. This is the order the layer

1183 predicts for the mean-only voxel, and the first test of that prediction on a
1184 body with genuine three-dimensional structure.

1185 What makes the sequence this clean is that the voxel carries a single error.
1186 The collocation voxel of the sharp sphere applied to this body, each cell at
1187 the profile's value at its centre and only the cells whose centres lie inside the
1188 sphere kept, carries two: its own internal-field error and the error of sampling
1189 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
1190 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
1191 2.0, 3.5 and 3.6 at $k_{\text{sa}} = 0.5$, and at $k_{\text{sa}} = 1$ the apparent orders are 1.7, 2.5
1192 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
1193 but its apparent order is not that of any single term and cannot be read from
1194 them. The same collocation voxel with the cell mean of the contrast, and
1195 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
1196 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
1197 second term, and with it removed the collocation and mean-only voxels differ
1198 only by their constants, as in Table 8. The cell mean, with every overlapping
1199 cell kept, represents the same body on every grid, and the order the layer
1200 predicts is then the one observed.

1201 The first-moment voxel of §8, extended to three dimensions with a con-
1202 trast linear in each cell, converges at fourth order on this same sphere (Ap-
1203 pendix Appendix E): its apparent orders are 4.01 and 4.13 where the dis-
1204 cretisation error is isolated at weak contrast, and at the full contrast its error
1205 with 16 voxels across is 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below
1206 the mean-only voxel's on the same grid.

1207 *The near field, an impedance march, and a sharp sphere.* Three further tests
 1208 are in Appendix Appendix D. Read at finite distance, down to a quarter
 1209 of a radius from the surface, the graded sphere converges at the orders of
 1210 the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-
 1211 moment voxel, while the asymptotic readout would be in error there by 48%
 1212 to 85% (Table D.18). Against the impedance march of Haines et al. [67],
 1213 which needs far fewer unknowns, the first-moment voxel reaches the march's
 1214 accuracy with six voxels across and continues at close to fourth order below
 1215 it, where the march is limited by the scattering between the spheres of the
 1216 periodic array it must solve (Table D.20). And a sphere with a sharp surface
 1217 does not converge: its error tracks the volume of its staircase, which changes
 1218 with the grid (Table D.21); the smooth sphere, like the layer, measures the
 1219 scheme.

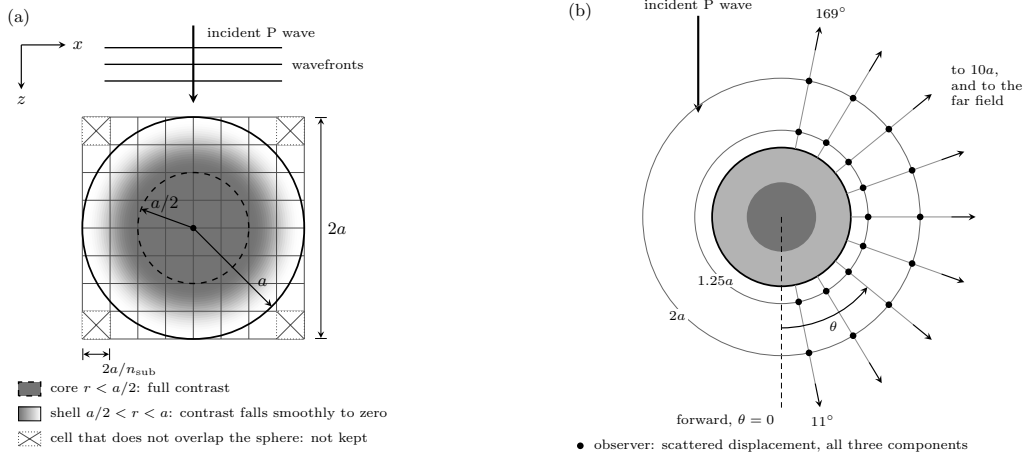


Figure 6: The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core $r < a/2$ and falls smoothly to zero across the shell $a/2 < r < a$. The circumscribing cube of side $2a$ is cut into n_{sub}^3 cells, here $n_{\text{sub}} = 8$; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along $+z$. (b) The observers. The scattered displacement is taken at nine scattering angles θ , measured from the forward direction and equally spaced from 11° to 169° , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

Alt text: Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

11. The Cartesian multipole hierarchy as a voxel scheme

The Legendre voxels of §8 represent the field of a voxel by its moments over the voxel. The hierarchy of §6 represents it by its gradients at the centre. Both are polynomial, and §6.4 shows that on the layer the hierarchy of degree $\ell = 3$ is fourth order, as the first-moment voxel is. This section builds the hierarchy into a voxel scheme in three dimensions and compares the two.

The scheme. Each voxel c , centred at $\mathbf{x}_c$, carries the coefficients $U_{j;W}^c$ of Eq. (20). The equation and its derivatives to order ℓ are imposed at every centre (with $\ell = 1$ and no density contrast this is the static numerical equivalent inclusion method on cuboidal elements, with uniform eigenstrains, equivalence at the element centres and a fast Fourier transform, of Zhou et al. [68]):

$$U_{i;P}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_P u_i^0(\mathbf{x}_c), \quad (46)$$

$$K_{in}[D; W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (47)$$

The two kinds of block are of different character. For $c' = c$ the integrand is singular and the integral is a distribution: it is the moment of the hierarchy, $K_{in}[D; W](0) = (-1)^{|D|} \mathbf{g}_{in}[D; W]$, in closed form on the cube. For $c' \neq c$ the field point is the centre of another voxel, at least half a voxel from the source voxel, so the integrand is smooth and a tensor Gauss rule integrates it; the derivatives of the Green's tensor, to any order, come from the family $x^a y^b z^c r^p e^{ikr}$, which is closed under differentiation. All the singular work of the scheme is thus in the self block, where it is done once and in closed

1242 form; the Legendre voxel's self block is a double integral of the same singular
1243 kernel. The blocks depend on the offset alone, so the product of the system
1244 with a vector is a convolution over the lattice and is done exactly by the fast
1245 Fourier transform. Where the contrast varies within a voxel it is projected
1246 there on polynomials of degree r , as in §9, and the source is the product of
1247 that polynomial and the field's: the moments then carry weights to degree
1248 $\ell + r$, and a block is a sum, over the monomials of the contrast, of blocks
1249 with the weight shifted.

1250 *Checks of the construction.* The derivatives of the Green's tensor to fourth
1251 order agree with the closed forms of its value, gradient and second derivative,
1252 and with central differences beyond them, to 8×10^{-7} . A lattice of one voxel
1253 reproduces the single site of §6.5 to 10^{-14} . Only the offsets of one octant
1254 are computed, the others following from the parity of each index; a block
1255 obtained that way equals one computed directly to 10^{-15} . The iterative
1256 solution equals a dense solution to 7×10^{-11} where both can be run.

1257 *A body with corners.* The cube of §6.5, at $k_{sh} = 0.2$, is cut into n^3 voxels.
1258 Table 11 gives the far-field error against the converged reference, with the
1259 Legendre voxels on the same body. On this body every degree of the hierarchy
1260 converges at about the same rate, an order near two between the two finest
1261 grids, and the degree buys a constant: thirty times from $\ell = 1$ to $\ell = 3$
1262 on 64 voxels. That rate is the body's and not the scheme's, for the mean-
1263 only Legendre voxel shows it too: a homogeneous cube has edges and corners,
1264 whose fields no polynomial follows. The uniform-strain closure and the mean-
1265 only voxel agree on one cell, as they must, being the same approximation.
1266 And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is

Table 11: A homogeneous cube cut into n^3 voxels, $k_S h = 0.2$: relative far-field error of the hierarchy of degree ℓ and of the Legendre voxels, with the unknowns of each voxel.

	unknowns a voxel	$n = 1$	2	3	4
hierarchy, $\ell = 1$	12	5.4×10^{-3}	1.7×10^{-3}	8.3×10^{-4}	5.0×10^{-4}
hierarchy, $\ell = 2$	30	1.6×10^{-3}	6.7×10^{-4}	3.7×10^{-4}	2.4×10^{-4}
hierarchy, $\ell = 3$	60	3.4×10^{-4}	7.0×10^{-5}	3.0×10^{-5}	1.7×10^{-5}
Legendre, mean only	9	5.4×10^{-3}	1.3×10^{-3}	5.8×10^{-4}	3.3×10^{-4}
Legendre, first moment	36	4.8×10^{-5}	6.0×10^{-6}		

more accurate than eight voxels of the hierarchy of degree three with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [54] established for the static equivalent inclusion method.

A smooth medium. The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §6.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile $\sin^2(\pi z/D)$ the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for $\ell = 1, 2$ and 3: a varying contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. The graded sphere of §10 has a shell

1283 5 m wide, and a solver that reaches fourteen voxels across puts fewer than
1284 four voxels across it; so the core is reduced here from $a/2$ to $a/10$, which
1285 widens the shell to 9 m, and a second profile, $\sin^2(\pi x/2)$ in the fraction x of
1286 the shell's width measured from the surface, is run beside the smoothstep.
1287 The exact reference is the radial solution of §10 for each body. The contrast
1288 is linear in each voxel for $\ell = 3$, which suffices for fourth order (§9), and of
degree ℓ for $\ell = 1$ and 2.

Table 12: The graded sphere with a shell 9 m wide, $k_{Sa} = 0.5$: relative far-field error
against the exact solution for n_{sub} voxels across, with the apparent order between the two
finest grids of each row. The lower block is the profile $\sin^2$.

	$n_{\text{sub}} = 4$	6	8	10	12	14	order
hierarchy, $\ell = 1$	1.5×10^{-3}	8.4×10^{-4}	4.9×10^{-4}	3.2×10^{-4}			1.94
hierarchy, $\ell = 2$	3.3×10^{-4}	2.1×10^{-4}	1.5×10^{-4}	1.0×10^{-4}			1.56
hierarchy, $\ell = 3$	2.6×10^{-4}	5.0×10^{-5}	1.7×10^{-5}	7.2×10^{-6}	3.6×10^{-6}	2.0×10^{-6}	3.80
Legendre, mean only	7.3×10^{-3}	3.2×10^{-3}	1.8×10^{-3}	1.1×10^{-3}			2.02
Legendre, first moment	2.9×10^{-4}	6.4×10^{-5}	2.2×10^{-5}	9.4×10^{-6}	4.6×10^{-6}	2.5×10^{-6}	3.92
hierarchy, $\ell = 3$	1.9×10^{-4}	3.8×10^{-5}	1.3×10^{-5}	5.2×10^{-6}	2.7×10^{-6}	1.5×10^{-6}	3.84
Legendre, first moment	2.2×10^{-4}	4.8×10^{-5}	1.6×10^{-5}	6.8×10^{-6}	3.4×10^{-6}	1.8×10^{-6}	4.02

1289
1290 Table 12 gives the result. The first-gradient scheme converges at second
1291 order, and is three to four times more accurate than the mean-only Legendre
1292 voxel on every grid, for twelve unknowns a voxel against nine. The second-
1293 gradient scheme is more accurate again but gains no order, as §6.4 predicts;
1294 its apparent order is still rising towards two. The third-gradient scheme
1295 converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8

1296 over the range from six to fourteen voxels across, for both profiles; the first-
 1297 moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over
 1298 the range. Neither has settled on four at fourteen voxels across, and the
 1299 second profile, gentler than the first, changes neither: the two fourth-order
 1300 schemes behave alike. On every grid from six voxels across the hierarchy's
 1301 error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel
 1302 against thirty-six, so that for a given number of unknowns the Legendre voxel
 1303 remains somewhat the more accurate; the wide margin of Table 11 was the
 1304 corners.

1305 *What the hierarchy is for.* The two schemes reach the same order by different
 1306 closures of the same equation, and they differ in what they cost and in what
 1307 they give besides. The Legendre voxel is the more economical in unknowns.
 1308 The hierarchy's blocks are simpler: its self block is known in closed form on
 1309 four constants (§6.2), and its coupling is a single integral of a smooth func-
 1310 tion where the Legendre voxel's is a double integral. The iteration converged
 1311 in five steps at every size; the time goes into the tables of blocks, which are
 1312 computed once for a grid and a frequency. Formed on the same footing, one
 1313 block for each orbit of offsets under the 48 symmetries of the cube, the others
 1314 following exactly, and every point value of the Green's tensor or its derivatives
 1315 from the same compiled radial ladder, the hierarchy's tables are the cheaper:
 1316 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and 23.3 s for
 1317 the first-moment Legendre voxel, with eight times fewer kernel evaluations.
 1318 Both have a one-off cost besides, the hierarchy's the smaller (0.3 s against
 1319 2.2 s): the moments of its self block depend on neither the frequency nor the
 1320 cell size, and are computed once and stored (`scripts/build_cube_unit_`

1321 `moments.py`; `scripts/measure_hierarchy_table_cost.py`; one machine,
1322 five threads). And the hierarchy is, in addition, a single-site T -matrix in
1323 closed form, whose accuracy improves block by block (§6.5), which no voxel
1324 defined by moments over the cell provides. The lattice results of this section
1325 come from one implementation, checked against exact solutions and, where
1326 it can be, against a dense solve; they are at one frequency and on one fam-
1327 ily of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_`
1328 `lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_`
1329 `hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.`
1330 `py`).

1331 12. Discussion

1332 Figure 7 collects the convergence of every test, with the order fitted to
1333 each by least squares (`scripts/plot_convergence_orders.py`).

1334 *What each basis is for.* A voxel holds two functions, and the results give
1335 each basis its part. The basis of the wavefield sets how well the voxel carries
1336 the field across itself: in a uniform medium the scheme converges at order
1337 $2p + 2$ with the constant c_p of Eq. (40), for P and S waves and at normal and
1338 oblique incidence alike. The basis of the medium sets how well the voxel *is*
1339 the medium: with degree r the order is capped at $2r + 2$ (Eq. (42)). Neither
1340 can make up for the other, and the second-order term shows why. The field
1341 scattered once is a copy of the medium’s profile, so the wavefield basis must
1342 carry the profile as well as the wave, and the error of that term is the relative
1343 projection error of the profile at the lower of the two degrees (Eq. (44)). At
1344 the frequencies of interest the wave varies little across a cell and the medium

1345 may vary much, so it is the medium, not the wavelength, that decides the
 1346 degree and the size of the cells. Two cases bracket the choice. Where the
 1347 medium is piecewise constant on the cells, as in a layered model with voxel
 1348 faces on its interfaces, a constant is exact for the medium, the projection
 1349 error vanishes, and the degree of the field is spent entirely on the wave.
 1350 Where the medium varies smoothly within the cells, the two degrees should
 1351 be equal: a higher degree for the contrast than for the field is wasted, and a
 1352 higher degree for the field improves only the Born term. That the two degrees
 1353 can be chosen independently is the practice of higher-order electromagnetic
 1354 volume methods [1]; what the present results add is the rule for choosing
 1355 them, and the error each choice leaves.

1356 *Where the error of a voxel scheme lives.* The results separate the two ap-
 1357 proximations a voxel scheme makes. The *coupling* between voxels makes
 1358 none: averaged over the source cell it is the continuum's cell integral exactly,
 1359 between planes by the form factor's zeros and within a plane by the tiling
 1360 identity, and the self term it omits is exactly the one the single site carries.
 1361 Every error therefore comes from the *single site*: from the internal field it
 1362 assumes and, where the medium varies within the cell, from the medium it
 1363 is given. Its T -matrix sums the rescattering within the voxel exactly, but
 1364 for the voxel as represented, and the split of the second-order term in §9
 1365 measures the consequence: the error sits in the part whose two scatterings
 1366 lie in one cell, and the part between cells carries at most 0.2% of it. For a
 1367 uniform field that error is the midpoint rule's, known in closed form; for a
 1368 linear field it is the next term of the same expansion. The converse is the
 1369 practical warning. A kernel that is not the cell integral (the point Green's

1370 tensor, or a cell average that does not match the single site’s closure) leaves
1371 a local term that does not vanish as the voxels shrink (§4): the evanescent
1372 lattice orders carry a static contribution of fixed size per voxel volume. A
1373 scheme built that way converges to the wrong limit, and refinement cannot
1374 reveal it, because the error is scale-invariant.

1375 *What the single site is.* The Cartesian multipole hierarchy of §6 turns the
1376 uniform-field single site from a modelling choice into a derived object with
1377 a stated error (§6.7). Its one departure from exact scattering by a sphere,
1378 the real $(ka)^2$ term of Eq. (F.1), and the collocation error of Eq. (36) have
1379 the same origin, the variation of the field across the scatterer, and the same
1380 remedy, a richer internal field. In a lattice the remedy is cheap: the field
1381 a voxel must represent is the smooth continuum field, not the field of an
1382 isolated cube, with its concentrations at the edges and corners.

1383 *Relation to other methods.* The partition of the error is the coupled-dipole
1384 method’s choice of polarisability seen from the other side: for space-filling
1385 cubes the tiling identity fixes the local field exactly, and the polarisability
1386 is neither needed nor a free parameter. The voxel of degree p is close kin
1387 to the spectral element of seismology; the two share a Galerkin condition
1388 on polynomial bases, and differ as a differential formulation from an integral
1389 one, and so suit different problems. Appendix Appendix H sets out both
1390 relations, and where each method is the better choice.

1391 *The tension between the two error sources.* The coupled-dipole method has
1392 two sources of error, the single site and the coupling between sites, and each
1393 has been corrected on its own: the polarisability by the Clausius–Mossotti

1394 relation, the lattice dispersion relation or the exact static limit, and the
 1395 coupling by integrating or filtering the Green tensor. The two are not in-
 1396 dependent. The point kernel's evanescent orders carry a static local term
 1397 whose size per voxel volume does not shrink with the voxels (§4), and the
 1398 single site must cancel exactly that term. A polarisability tuned for one
 1399 kernel is therefore wrong for another: the lattice dispersion relation corrects
 1400 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 1401 tered one changes the term it must cancel. Tuning either alone moves the
 1402 error between the two, and refinement cannot reveal a mismatch, because
 1403 the surviving term is scale-invariant. For space-filling cubes the tension is
 1404 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 1405 term that the closure carries (Eq. (35)), so the coupling error vanishes iden-
 1406 tically and every remaining error is the single site's, the variation of the field
 1407 across the cell. That error has a second tension of its own. It is a midpoint-
 1408 rule error whose constants have opposite signs in reflection and transmission
 1409 (Eq. (36)), so any scalar adjustment of the single site that improves one wors-
 1410 ens the other; the package's far-field corrections reduce the reflection error
 1411 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 1412 however chosen, can remove it. What removes it is a richer internal field:
 1413 the first moment lowers the error by two orders in kd in both observables at
 1414 once.

1415 *The cost of higher order, and a fast solver.* A first-moment voxel carries four
 1416 times as many unknowns as a uniform one, but by the closed-form constants it
 1417 reaches a reflection error of 10^{-6} with voxels 58 times larger in each direction;
 1418 the stratified model of §9 measures it directly, 347 planes of uniform-field

1419 voxels against 8 first-moment voxels. And because the discretisation error
 1420 is known in advance and the coupling between cells is exact, a fast product
 1421 need be no more accurate than that error, and can be budgeted separately
 1422 from it. Appendix Appendix I sets out the cost, and what the error analysis
 1423 offers a fast solver.

1424 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 1425 mogeneous layer and a randomly stratified one, for which the discrete system
 1426 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 1427 moment voxel in a full three-dimensional solver, where the gradient channels
 1428 couple through the evanescent lattice orders (§8) and the tiling identity holds
 1429 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 1430 voxel converging at second order on a laterally heterogeneous body against
 1431 an exact solution, and the first-moment voxel converges at fourth order there
 1432 (§11). A contrast that varies within a cell, which §9 shows is needed beyond
 1433 second order in a smoothly varying medium, has a further consequence in
 1434 three dimensions: its linear part is odd about the cell’s centre, so it couples
 1435 the single site’s uniform-field and first-moment blocks, which parity keeps
 1436 apart for a homogeneous cube (§6.3). A graded voxel therefore needs at least
 1437 the first-moment block; that voxel is built, and tested on the graded sphere,
 1438 in the companion paper. The numerical evidence is at moderate, physically
 1439 typical contrasts (up to 15% in density and moduli) and at $k_S h \leq 0.3$, the
 1440 range in which the single-site T -matrix is validated; the closed forms are first
 1441 order in the contrast, and the departures from them at the full contrast are of
 1442 second order. What is established is the *order* of convergence and its leading
 1443 constant, not the behaviour at strong contrast, where the single site’s own

1444 closure may limit the accuracy before the lattice coupling does. The single
 1445 site's dynamic error is in closed form for a sphere only; for a cube, for which
 1446 no exact solution exists, it is known through the layer (§§7–8) and, for the
 1447 isolated cube, by comparison with a converged solution (§6.5). The hier-
 1448 archy beyond the first gradient is established less widely than the rest: its
 1449 orders and its error factor on the layer are derived symbolically and checked
 1450 independently to the third degree, and its moments are confirmed by two
 1451 routes, but its three-dimensional results, the single site against the sphere
 1452 and the cube and the voxel scheme of §11, come from one implementation, at
 1453 one frequency for the voxel scheme, checked against exact solutions. There
 1454 the fourth order is approached and not attained on the grids reached (3.8
 1455 at fourteen voxels across, as for the first-moment voxel), the fifth-gradient
 1456 system, whose moments are in hand, has not been assembled in three di-
 1457 mensions, and the two fourth-order schemes have been compared in the cost
 1458 of forming their tables only (§11). Two extensions follow directly. The first
 1459 is the three-dimensional solver itself. The second is a stratified *background*,
 1460 a layered reference medium, as distinct from the stratified contrast of §9.
 1461 For a laterally uniform layer of voxels the orders, the first-order factor and
 1462 the projection-error law all survive it, at normal and at oblique incidence,
 1463 for P, SV and SH waves, in a background of two half-spaces and in one of
 1464 three layers (§9). What remains is a body of finite lateral extent in such a
 1465 background, where the image of a voxel in an interface it touches is singular.

13. Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree p and the contrast to degree r in each voxel, the scheme converges at order $\min(2p+2, 2r+2)$, with the leading error $c_p(kd)^{2p+2}$ in closed form (Eq. (40)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree $\min(p, r)$: it is the single site's, whose T -matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a Cartesian multipole hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [53] to bodies that are not ellipsoids, whose

1491 moments are in closed form on four constants. Its blocks gain accuracy
1492 in pairs, with the error of each truncation in closed form: as a single site
1493 the third-gradient system is three hundred times more accurate than the
1494 uniform-strain closure against the exact sphere, and as a voxel scheme it
1495 is fourth order, as the first-moment voxel is, on a smoothly graded sphere,
1496 where in a lattice the Legendre voxel is the more economical. The results
1497 and the evidence for each are collected in Appendix Appendix J.

1498 *The error is known in advance.* For every discretisation studied here the
1499 error is not something to be discovered by refining until the answer stops
1500 changing. Its order and its leading constant are known before anything is
1501 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
1502 collocation voxel (Eq. (36)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
1503 voxels of degree 0, 1 and 2 (Eq. (40)), and the order $\min(2p + 2, 2r + 2)$
1504 where the medium varies within a cell (Eq. (42)), with the size of the error
1505 there given by the relative projection error of the medium (Eq. (44)), which
1506 is computed from the model and the grid alone. That is worth more than
1507 the order alone, for three reasons. First, the cell size and the degree of the
1508 voxel that a required accuracy needs follow from the wavenumber and the
1509 medium, before the computation and without a refinement study. Secondly,
1510 it turns a refinement study from a hope into a test. A scheme of this kind can
1511 converge smoothly to the wrong limit (§4), and smooth convergence alone
1512 cannot reveal it; but the difference between two grids must have the known
1513 order and the known constant, and a computation whose differences do not
1514 is in error from some cause other than the discretisation. Thirdly, it shows
1515 where effort pays. The error is that of representing the field and the medium

1516 within a cell, so raising the degree of the cell does what multiplying the
 1517 cells cannot: in the stratified model one first-moment voxel per layer reaches
 1518 10^{-6} where the uniform-field voxel needs about forty. Whether the error of
 1519 a body with three-dimensional structure can be known in advance in the
 1520 same way is the question taken up in a following paper [69], where it decides
 1521 where the cells of an adaptive grid are placed. That paper continues the
 1522 paper's first strand, the two bases and their error, with the Legendre voxels.
 1523 The second strand, the Cartesian multipole hierarchy, whose coefficients are
 1524 integrals of the Green's tensor over the cube, is taken further in a paper on
 1525 the coupling integrals of polynomial cubes in closed form, of which it is the
 1526 worked application.

1527 **Appendix A. The moment engine: gates and a second route**

1528 Two checks, whose answers do not depend on the engine, gate every grade.
 1529 Because $\nabla^2 r^m = m(m+1)r^{m-2}$ and $\nabla^2(1/r) = -4\pi\delta$,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi (-1)^{|R|} (\partial_R x_W)(0),$$

(A.1)

1530 for any further derivatives R and any weight W . The second is the sum rule
 1531 (27) at every grade: it tests the δ function of the inclusion problem and its
 1532 derivatives, which is where a quadrature fails. Both hold for every grade of
 1533 Table 2 with two or more derivatives, 910 traces of each kind; the grades with
 1534 fewer, for which the integrand is integrable, agree with numerical integration
 1535 over the cube.

1536 The closed forms are also confirmed by a route that shares nothing with
 1537 the faces of the cube. The cube is its inscribed ball plus the remainder.

1538 On the ball the derivatives are moved onto the sphere, where the surface
 1539 integrals of monomials are Gamma functions, and all the distributional con-
 1540 tent is there; on the remainder the integrand is an ordinary function, in-
 1541 tegrated radially in closed form and over directions through the six faces
 1542 by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored
 1543 values, the largest difference being 1.1×10^{-12} , and satisfies Eq. (A.1) by
 1544 itself to 5×10^{-10} or better up to eight derivatives (`CubeMomentCore.wl`,
 1545 `CubeMomentHigherGrades.wl`, `CubeMomentStore.wl`, `CubeMomentCanonical.`
 1546 `wl`; `scripts/crosscheck_cube_moments_ball_shell.py`).

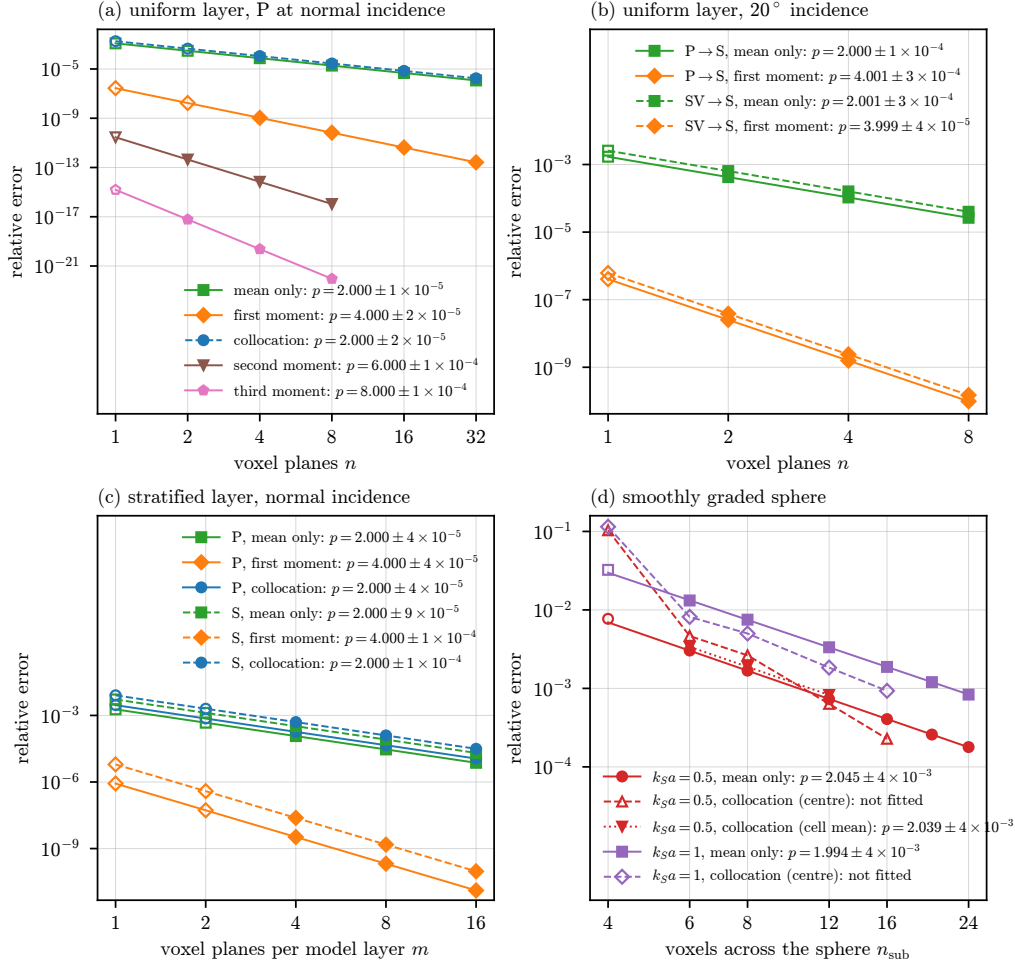


Figure 7: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (38) and (B.1)). (a) The uniform layer, P reflection at normal incidence ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 8; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and

1547 Appendix B. Further tests of the Legendre voxel

1548 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
 1549 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (\text{B.1})$$

1550 with L_a the lateral Legendre factors. The whole-space tensor decays as
 1551 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
 1552 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma +$
 1553 $\mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
 1554 averaging makes the lattice sum converge without Ewald splitting, to three
 1555 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
 and 40° (`ContinuumLimit_Oblique.wl`; Table B.13). The mean-only voxel

Table B.13: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (B.1). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

1556
 1557 is second order and the first-moment voxel fourth order in both the reflected
 1558 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$

1559 and the evanescent coupling of the gradient channels do not cost an order.

1560 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (38) to all printed digits.

1561 *Incident S.* Nothing in the construction is specific to P. For a normally in-
 1562 cident S wave the layer is the one-dimensional problem of Eq. (38) with
 1563 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
 1564 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
 1565 transmission, and the full 9×9 construction with an incident SH wave re-
 1566 produces that one-dimensional result to all printed digits. At oblique inci-
 1567 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
 are those of Table B.14. The orders, and with $k = k_S$ the closed-form con-

Table B.14: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

1568

1569 stants of Eqs. (36) and (39), are those of the incident P wave.

1570 *Degree two at oblique incidence.* In three dimensions the voxel of degree
 1571 2 carries, on each of the nine components of its state, the ten Legendre
 1572 products of total degree at most two in (z, x, y) . In the construction of
 1573 Eq. (B.1) it converges at sixth order against the exact layer for incident
 1574 P, SV and SH waves (Table B.15; `scripts/measure_oblique_degree.py`).

1575 The test is run at $\omega = 3000 \text{ rad s}^{-1}$ ($k_P D = 1.2$, $k_S D = 2$), ten times the
1576 frequency of the other tables, because at 300 rad s^{-1} a sixth-order error on
1577 these grids lies below the round-off of double precision. The exact reflection
1578 coefficient is that of Kennett’s recursion, an independent algorithm, and the
1579 same computation with the bases of degree 0 and 1 returns orders 2 and 4
1580 in all five cases. At normal incidence the error on four planes, 6.95×10^{-9} , is
1581 the $c_2(kd)^6 = 7.2 \times 10^{-9}$ of Eq. (40) to 4%. Doubling the number of lattice
1582 orders retained in Eq. (B.1) changes the errors by less than 1%. Unlike the
1583 degrees 0 and 1, this degree is implemented once at oblique incidence; its
check is the exact solution.

Table B.15: The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000 \text{ rad s}^{-1}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

incident wave	error, $p = 2$			order	orders
	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0 / p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20° : R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20° : R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20° : R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40° : R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

1584

1585 The spectral construction is implemented twice. The second implementa-
1586 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
1587 nition of the scheme: it takes the depth dependence of the transformed kernel
1588 from an independent residue construction, obtains the local weight as a nu-

1589 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
 1590 ble integral by quadrature and separates P from S by sampling the upgoing
 1591 field at two depths. The raw reflected displacements of the two agree to $9 \times$
 1592 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
 1593 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
 1594 scheme of any degree is also implemented twice. The second implementation
 1595 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
 1596 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
 1597 from its own interface system, in double precision. Its errors agree with the
 1598 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 1599 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 1600 6.00; for one plane as the layer thins its differences from the exact layer fall
 1601 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 1602 Figure 7(a,b) shows the convergence, with the orders fitted to it.

1603 **Appendix C. The stratified-background construction**

1604 The experiment is built in four steps.

1605 *The kernel.* At each lateral wavenumber $\boldsymbol{\kappa}$ the whole-space kernel between
 1606 two depths factorises over the plane-wave modes, three going down and three
 1607 going up (P, SV, SH): $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') = \mathbf{D}_\downarrow \mathbf{E}(z - z') \mathbf{S}_\downarrow$ for $z > z'$, with $\mathbf{S}_\downarrow$ the
 1608 3×9 matrix of mode amplitudes radiated downwards by a nine-component
 1609 source, $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$ the propagation over a distance L with γ_m the
 1610 vertical wavenumber of mode m , and $\mathbf{D}_\downarrow$ the 9×3 matrix of the modes'
 1611 states; and likewise upwards. With the interfaces at z_A above and z_B below,
 1612 $H = z_B - z_A$, and $\mathbf{R}_A, \mathbf{R}_B$ the 3×3 reflection matrices of the two interfaces

1613 seen from inside the layer, a source at z' sends $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow q$ down and $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow q$
 1614 up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\boldsymbol{\kappa}; z, z') q &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \tag{C.1}$$

1615 with $\mathbf{d}_A$ the downgoing amplitudes leaving the upper interface and $\mathbf{u}_B$ the
 1616 upgoing ones leaving the lower. It is added to the whole-space kernel in
 1617 every block of Eq. (B.1), at every order of the reciprocal lattice. It is smooth
 1618 and separable in z and z' , so its integrals over the cells are one-dimensional
 1619 moments of exponentials. For two half-spaces $\mathbf{R}_A = 0$ and only the second
 1620 line survives.

1621 *The interfaces.* $\mathbf{R}_A$, $\mathbf{R}_B$ and the transmission matrices into and out of the
 1622 layer follow from continuity of displacement and traction at each interface:
 1623 six equations in the states of the six modes on either side, solved for each
 1624 lateral wavenumber, evanescent orders included.

1625 *The incident field and the output.* A plane wave of unit amplitude arrives
 1626 from the upper half-space. Its field in the layer, a downgoing part and an
 1627 upgoing part with the reverberation summed as in Eq. (C.1), is the right-
 1628 hand side of the scheme. The output is the upgoing wave of the same type
 1629 as the incident one, arriving at z_A from the voxels and transmitted into the
 1630 upper half-space: the change that the layer of voxels makes to the same-type
 1631 reflection coefficient (R_{PP} , R_{SS} or R_{SH}), a quantity that does not depend on
 1632 how the waves are normalised.

1633 *The reference, and a check of the construction.* The exact change is the

1634 same-type reflection coefficient of the stack with the layer, less that of the
 1635 background, both from Kennett’s recursion [65], an independent algorithm.
 1636 Before any voxel is introduced, the background’s own reflection coefficient
 1637 computed from the modes and the interface matrices above agrees with Ken-
 1638 nett’s to 5×10^{-15} in all twelve cases.

1639 It is computed by two independent algorithms. The first is invariant
 1640 imbedding, the impedance march of Haines et al. [67]: the six-component
 1641 state of displacement and traction obeys $dy/dz = A(z)y$, and with the four
 1642 3×3 blocks A_{ij} of A the impedance Y , which gives the traction from the
 1643 displacement, obeys the Riccati equation

$$\frac{dY}{dz} = A_{21} + A_{22}Y - YA_{11} - YA_{12}Y. \quad (C.2)$$

1644 It is started below the stack with the impedance of the downgoing waves
 1645 of the lower half-space, which is the radiation condition, marched upwards,
 1646 exactly through the homogeneous gaps and by fourth-order Runge–Kutta
 1647 with the medium sampled at the stages through the graded layer, and at
 1648 the top the reflection matrix follows from continuity of displacement and
 1649 traction with the modes of the upper half-space. Y stays bounded, so no
 1650 growing exponential is formed. The second is the propagator: the same
 1651 state is integrated across the graded layer as a 6×6 matrix, with $A(z) =$
 1652 $F(z)\text{diag}(i\gamma_m)F(z)^{-1}$ built at each depth from the plane-wave modes of the
 1653 medium there, F the matrix of the six modes’ displacements and tractions
 1654 and γ_m their vertical wavenumbers, and the radiation conditions of the two
 1655 half-spaces are imposed on the result. For the graded layer the two agree
 1656 to 10^{-12} in all twelve cases with 400 steps of the march, whose error falls
 1657 sixteenfold for each doubling of the steps; for a uniform layer the propagator

1658 equals Kennett's recursion to 3×10^{-14} .

1659 *Oblique incidence, and a three-layer background.* At oblique incidence the
1660 test is the three-dimensional scheme of Eq. (B.1), with P, SV and SH waves,
1661 in two backgrounds: the two half-spaces above, and a three-layer background
1662 in which the voxels lie in a layer between a slower half-space above ($\alpha = 4200$,
1663 $\beta = 2400 \text{ m s}^{-1}$, $\rho = 2300 \text{ kg m}^{-3}$) and the faster one below (6500, 3700,
1664 2800), so that the layer reverberates between two interfaces. The construc-
1665 tion is the one set out above: the reverberation added to the whole-space
1666 kernel at every order of the reciprocal lattice, Eq. (C.1), the interface ma-
1667 trices, the incident field, and the reference, Kennett's recursion [65], against
1668 which the background's own reflection coefficient is reproduced to 5×10^{-15}
1669 in all twelve cases before any voxel is introduced.

1670 The results are in Table C.16, at $\omega = 3000 \text{ rad s}^{-1}$ ($k_S D = 2$; at 300 rad s^{-1}
1671 the sixth-order errors lie below round-off, as in Table B.15) and 20° in the
1672 upper half-space. In all thirty-six cases the order is that of the whole space,
1673 with the interfaces 3m from the layer and with the voxels against them
1674 (`scripts/measure_oblique_stratified_background.py`). The errors are
1675 close to those of the whole space at the same frequency (Table B.15): the
1676 background changes the field the voxels must represent, not how well they
1677 represent it.

1678 *The two terms of the error at oblique incidence.* The orders say that the
1679 scheme converges; the two-term analysis says what its error is. Both terms
1680 are tested at 20° , in the whole space and in the stratified backgrounds, for
1681 P, SV and SH waves.

Table C.16: The layer of voxels in a stratified background at 20° , $\omega = 3000 \text{ rad s}^{-1}$: relative error of the change of the same-type reflection coefficient on $n = 4$ planes, and the order over the last halving ($n = 2 \rightarrow 4$), for field degrees $p = 0, 1$ and 2 .

background	wave	$p = 0$	$p = 1$	$p = 2$
two half-spaces, interface 3 m below	P	6.1×10^{-3} (2.03)	2.7×10^{-6} (4.02)	2.3×10^{-9} (6.04)
	SV	1.5×10^{-2} (2.11)	2.6×10^{-5} (4.13)	3.7×10^{-8} (6.09)
	SH	1.6×10^{-2} (2.11)	2.7×10^{-5} (4.13)	4.1×10^{-8} (6.09)
two half-spaces, interface against the voxels	P	5.1×10^{-3} (2.04)	3.8×10^{-6} (4.04)	2.1×10^{-9} (6.00)
	SV	1.6×10^{-2} (2.10)	2.5×10^{-5} (4.14)	3.3×10^{-8} (6.11)
	SH	1.7×10^{-2} (2.10)	2.4×10^{-5} (4.15)	3.4×10^{-8} (6.11)
three layers, interfaces 3 m away	P	6.3×10^{-3} (2.03)	8.7×10^{-7} (4.04)	2.4×10^{-9} (6.06)
	SV	1.0×10^{-2} (2.10)	5.8×10^{-6} (4.08)	5.2×10^{-8} (6.11)
	SH	1.3×10^{-2} (2.10)	3.0×10^{-6} (3.95)	9.3×10^{-8} (6.10)
three layers, interfaces against the voxels	P	4.8×10^{-3} (2.04)	2.4×10^{-6} (4.04)	4.1×10^{-9} (6.03)
	SV	8.5×10^{-3} (2.09)	7.1×10^{-6} (4.01)	5.0×10^{-8} (6.09)
	SH	1.4×10^{-2} (2.10)	5.8×10^{-6} (3.92)	8.5×10^{-8} (6.12)

1682 *The first-order term.* The scheme is written as $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$ with the
1683 output $u = \mathbf{r} \mathbf{c}$, as in Eq. (G.8), so that $T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}$ with no differencing.
1684 The prediction is built from plane waves. The background field in the layer
1685 is a sum of six, P, SV and SH going down and going up, and the voxels
1686 radiate into six. For a pair made of an incident wave of wavevector $\mathbf{k}_{\text{in}}$ and
1687 a radiated wave of wavevector $\mathbf{k}_{\text{out}}$, a cubic voxel of half-side h whose field is
1688 held to total degree p gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (\text{C.3})$$

1689 the sum being over the Legendre products $P_{a_1} P_{a_2} P_{a_3}$ of the voxel's basis and
1690 j_n the spherical Bessel functions: the numerator is the projection of the inci-
1691 dent wave onto the basis paired with that of the outgoing wave, the denomi-
1692 nator the exact pairing. Its leading departure from one is $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$ for
1693 $p = 0$, with $d = 2h$, and along one axis it is the factor of Eq. (39). The pre-
1694 dicted first-order term is the sum over the thirty-six pairs of the exact term
1695 of each pair, an integral of two exponentials over the voxel, times its factor,
1696 the radiated waves then being carried to the observer by the reverberation
1697 of Eq. (C.1) exactly. On one and two planes of voxels of degree 0, 1 and
1698 2, at $\omega = 3000 \text{ rad s}^{-1}$, the scheme's first-order term equals this prediction
1699 to 2×10^{-15} in all twelve combinations of background and wave type. The
1700 first-order error is therefore known in closed form at oblique incidence, in a
1701 stratified background, for every wave type.

1702 *The second-order term.* For the law of Eq. (44) the contrast must vary

1703 within the voxels. Each plane of voxels is given the smooth profile of §9
 1704 in depth, projected to Legendre degree $r = p$, and the coupling is that of
 1705 Eq. (41) with the reverberation added. The second-order term of the scheme
 1706 is $T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}$. The exact T_2 needs the exact stack with a layer
 1707 whose properties vary continuously in depth, at oblique incidence, for which
 1708 a recursion over homogeneous layers is not exact. It is computed by two
 1709 independent algorithms, invariant imbedding [67] and the propagator of the
 1710 six-component state, which agree to 10^{-12} in all twelve cases (above). T_2 then
 1711 follows from Eq. (G.9) at $\epsilon = 0.1$, where halving ϵ changes it by 2×10^{-8} .
 1712 The ratio of the relative error of T_2 to the relative projection error $\mathcal{E}_p$ is in
 1713 Table C.17. On four planes the law holds to 3% for $p = 0$ and 1 and to 6%
 1714 for $p = 2$ at $k_S D = 0.2$, and to 0.6% for all three degrees at $k_S D = 0.067$, in
 1715 every background and for every wave type: the departure falls about ninefold
 1716 as the frequency falls threefold, as $(kD)^2$. On two planes the ratios for $p = 0$
 1717 and 1 lie within 0.03 of those in the table; for $p = 2$ they are lower, 0.89 to 0.99
 1718 at $k_S D = 0.2$ and 0.988 to 0.999 at $k_S D = 0.067$. The law is loosest for the
 1719 highest degree at the highest frequency, because the projection error of degree
 1720 2 is so small (1.6×10^{-5} on four planes) that the departure from the long-wave
 1721 limit is a larger share of it. Both terms of the error are thus known in advance
 1722 at oblique incidence and in a stratified background, as they are at normal
 1723 incidence in the whole space (`scripts/measure_oblique_two_term.py`).

1724 Two limits stand. The oblique construction is implemented once, and
 1725 its checks are the exact solutions. And the voxels here fill their plane, so
 1726 that the image of a voxel in an interface it touches is the image of a whole
 1727 plane; a body of finite lateral extent against an interface, where that image

Table C.17: The relative error of the second-order term over the relative projection error of the medium, at 20° , on $n = 4$ planes with $r = p$; the law of Eq. (44) predicts one in the long-wave limit.

		$k_S D = 0.2$			$k_S D = 0.067$		
background	wave	$p = 0$	$p = 1$	$p = 2$	$p = 0$	$p = 1$	$p = 2$
whole space	P	1.000	0.998	1.011	1.000	1.000	1.000
	SV	0.998	0.992	1.032	1.000	0.999	1.004
	SH	0.999	0.992	1.031	1.000	0.999	1.003
two half-spaces, interface 3 m below	P	1.010	1.009	1.019	1.001	1.001	1.001
	SV	1.012	1.008	1.046	1.001	1.001	1.005
	SH	1.027	1.025	1.059	1.003	1.003	1.006
three layers, interfaces 3 m away	P	1.004	1.003	1.013	1.000	1.000	1.000
	SV	0.979	0.973	1.002	0.997	0.996	1.000
	SH	1.007	1.004	1.035	1.001	1.001	1.004
three layers, interfaces against the voxels	P	1.002	1.001	1.011	1.000	1.000	1.000
	SV	0.995	0.988	1.018	0.999	0.999	1.002
	SH	1.004	1.002	1.032	1.000	1.000	1.003

1728 is singular, has not been run.

1729 **Appendix D. The sphere: near field, an impedance march and a** 1730 **sharp surface**

1731 *The near field.* A receiver is seldom at infinity, and the scheme does not
1732 require it to be. The scattered field at a point $\mathbf{x}$ outside the voxels is the
1733 propagator applied to the source that each voxel carries, the contrast times
1734 the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \boldsymbol{\Delta}(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (\text{D.1})$$

1735 with $\mathbf{K}$ the nine-component Green's tensor of §2, the same one that couples
1736 the voxels in the solve, $\boldsymbol{\Delta}$ the contrast operator of Eq. (5) and e_j the poly-
1737 nomial internal state of voxel j . Its first three rows are the displacement and
1738 its last six the strain. Nothing is dropped: the terms in $1/r^2$ and $1/r^3$ are in
1739 $\mathbf{K}$. The integrand's source is a polynomial in each voxel and is integrated by
1740 a tensor Gauss rule with six nodes a side; the rule's own error is measured
1741 by repeating the readout with four nodes, and from 8 voxels across the two
1742 differ by less than 4×10^{-6} at $1.25a$ and by less than 10^{-10} at $2a$. The ob-
1743 servers are the nine angles of Fig. 6(b) on spheres of radius $1.25a$, $2a$ and
1744 $10a$, where $k_S r$ runs from 0.6 to 10; the reference is the exact partial-wave
1745 displacement at the same points; and the error is the measure of the far-field
1746 test. Table D.18 gives it for the mean-only voxel and for the first-moment
1747 voxel (Appendix Appendix E). The orders of the far field hold in the near
1748 field: second for the mean-only voxel at every distance and both frequencies,
1749 and 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the

Table D.18: The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

n_{sub}	mean only			first moment		
	1.25a	2a	10a	1.25a	2a	10a
$k_S a = 0.5$						
4	3.3×10^{-2}	1.2×10^{-2}	5.3×10^{-3}	1.4×10^{-2}	1.6×10^{-3}	6.3×10^{-4}
6	1.4×10^{-2}	5.2×10^{-3}	1.9×10^{-3}	1.2×10^{-3}	3.8×10^{-4}	1.9×10^{-4}
8	8.0×10^{-3}	2.9×10^{-3}	1.1×10^{-3}	4.7×10^{-4}	1.4×10^{-4}	7.3×10^{-5}
12	3.5×10^{-3}	1.2×10^{-3}	4.7×10^{-4}	7.8×10^{-5}	3.1×10^{-5}	1.7×10^{-5}
16	2.0×10^{-3}	6.7×10^{-4}	2.6×10^{-4}	2.5×10^{-5}	1.0×10^{-5}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.03	2.06	2.02	3.95	3.85	3.77
$k_S a = 1.0$						
4	4.8×10^{-2}	3.5×10^{-2}	3.3×10^{-2}	9.9×10^{-3}	1.3×10^{-3}	6.0×10^{-4}
6	2.0×10^{-2}	1.4×10^{-2}	1.4×10^{-2}	9.3×10^{-4}	3.0×10^{-4}	2.0×10^{-4}
8	1.2×10^{-2}	8.1×10^{-3}	7.8×10^{-3}	3.5×10^{-4}	1.1×10^{-4}	7.4×10^{-5}
12	5.2×10^{-3}	3.6×10^{-3}	3.5×10^{-3}	6.0×10^{-5}	2.5×10^{-5}	1.7×10^{-5}
16	2.9×10^{-3}	2.0×10^{-3}	1.9×10^{-3}	1.9×10^{-5}	8.3×10^{-6}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.01	2.01	2.00	3.95	3.84	3.78

1750 far field at this contrast. A quarter of a radius from the surface, with 16 vox-
 1751 els across, the first-moment voxel's error is 2.5×10^{-5} and 1.9×10^{-5} , 78 and
 1752 153 times below the mean-only voxel's. The error grows towards the body,
 1753 by a factor of 1.5 to 7 between $10a$ and $1.25a$, because the near field carries
 1754 the higher multipoles of the source, which a given grid resolves less well; the
 1755 order does not change. On the coarsest grid a voxel is as wide as the graded
 1756 shell and twice as wide as the gap to the nearest observer, and the sequence
 1757 is not yet asymptotic. Read at these distances with the asymptotic formula,
 1758 the same solutions would be in error by 48% to 85%: the far-field readout
 1759 is for the far field (`scripts/measure_graded_sphere_near_field.py`; the
 1760 readout is `graded_field` of the package, tested against the propagator of a
 1761 point source, against the gradient of its own displacement, and against the
 1762 far-field readout at great distance).

1763 *The voxels against an impedance march.* A second route to the same body
 1764 is invariant imbedding. The impedance of §9, Eq. (C.2), is marched up-
 1765 wards through the depth range of the sphere with the lateral variation of the
 1766 medium carried inside the march, on a laterally periodic grid, and a second
 1767 sweep carries the field down [67]; the march returns the reflected plane waves
 1768 on a plane above the body and the transmitted ones on a plane below. It is
 1769 run here on the graded sphere and set beside the voxels, to ask how many
 1770 voxels a given accuracy costs against an established alternative. Four things
 1771 are fixed so that the comparison is of like with like.

- 1772 1. *The planes.* Both methods are read on the planes $z = -1.25a$ and
 1773 $z = +1.25a$. On the tangent planes $z = \pm a$ themselves the observers lie
 1774 on the faces of the voxels at the poles, where the Gauss rule of the voxel

Table D.19: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

1775 readout (Eq. (D.1)) cannot integrate the singular kernel: its change
1776 from four to six nodes is 2% to 10% there, more than the scheme's error,
1777 and below 10^{-4} a quarter of a radius away. The march's amplitudes are
1778 carried across the same homogeneous gap by each order's own phase,
1779 which is exact.

1780 2. *The periodic array.* The march needs a laterally periodic grid, so it
1781 solves a square array of spheres. Its answer is scored against the exact
1782 solution of one sphere summed over the array, by Poisson summation,
1783 which is exact but for scattering between the spheres. That part, and
1784 the lateral resolution, are errors of the method as used on a single body,
1785 and are counted.

1786 3. *The error.* For the march, the norm of the difference over the plane-

1787 wave orders of the grid, relative to the norm of the exact scattered
1788 orders; for the voxels, the norm of the difference of the scattered dis-
1789 placement over a patch of side $5a$ on each plane, relative to the norm
1790 of the exact one. The two are the same quantity in wavenumber and
1791 in space, for the array and for the single body.

1792 4. *The refinement.* For the march, the lateral grid, with the depth step
1793 refined until the error no longer changes; for the voxels, the grid and
1794 the degree.

Table D.20: The graded sphere at $k_S a = 0.5$ read on the planes $z = \mp 1.25a$: relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is $5a$ unless stated.

method	setting	unknowns	reflection	transmission
impedance march	12×12 , 16 steps	432	1.2×10^{-2}	9.9×10^{-3}
	16×16 , 32 steps	768	4.3×10^{-3}	3.6×10^{-3}
	20×20 , 32 steps	1200	7.6×10^{-4}	7.3×10^{-4}
	28×28 , 48 steps	2352	2.5×10^{-4}	5.6×10^{-4}
	30×30 , period $7.5a$	2700	7.2×10^{-4}	4.7×10^{-4}
mean-only voxel	8 across	3672	5.0×10^{-3}	7.9×10^{-3}
	16 across	24 984	1.2×10^{-3}	2.0×10^{-3}
first-moment voxel	6 across	7488	3.3×10^{-4}	5.2×10^{-4}
	8 across	14 688	1.4×10^{-4}	2.2×10^{-4}
	16 across	99 936	1.1×10^{-5}	1.8×10^{-5}
second-moment voxel	8 across	36 720	1.4×10^{-5}	2.3×10^{-5}
	10 across	69 120	2.6×10^{-6}	4.2×10^{-6}

1795 Table D.20 gives the result. The march converges quickly in the lateral grid,

1796 because the body is smooth, and with far fewer unknowns than any voxel
 1797 scheme: at 20×20 it is at 7×10^{-4} with 1200 unknowns on a plane. The
 1798 mean-only voxel does not reach that with sixteen voxels across and twenty
 1799 times the unknowns; on a sharp sphere, where it also carries a staircase, it
 1800 fares worse. The first-moment voxel reaches the march's accuracy with six
 1801 voxels across, and below about 2×10^{-4} it continues at close to fourth order
 1802 where the march does not: in transmission the march's error at a period
 1803 of $5a$ stops near 5×10^{-4} , which is the scattering between the spheres of
 1804 its array, since lengthening the period lowers it and refining the grid does
 1805 not. The costs are of different kinds. The march's unknowns are few but its
 1806 operator is a dense matrix on the plane, marched through the depth, and a
 1807 finer lateral grid requires a smaller depth step (with eight steps the 20×20
 1808 march diverged); the voxel schemes have many unknowns and a product done
 1809 by fast Fourier transform. In the timings of these runs, taken on one machine
 1810 with other work in progress and therefore only indicative, the first-moment
 1811 voxel at six across was solved in about a sixth of the time of the 20×20 march.
 1812 The march has what the comparison does not use: it returns the reflection
 1813 and transmission for every incident plane wave of the grid at once, and a
 1814 laterally periodic or laterally extended medium is its natural problem and
 1815 not a cost. The comparison is at one frequency (`scripts/measure_graded_`
 1816 `sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

1817 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 1818 and its voxelisation changes with every grid. Table D.21 shows its error for
 1819 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_S a = 0.5$.
 1820 The error tracks the staircase's volume, which is 5%, 17% and 4% too large

1821 on these grids. Rescaled by the ratio of the sphere’s volume to the staircase’s,
1822 it falls below 1% at $k_S a = 0.5$ and to at most 3% at $k_S a = 1.0$, but it still
1823 does not converge: what remains is the staircase’s shape. A refinement study
1824 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
1825 measures the scheme.

Table D.21: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_S a = 0.5$		$k_S a = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

1826 Appendix E. The three-dimensional voxel solver

1827 The three-dimensional results of §§10 and 11 use the Legendre voxels
1828 of §8 on a cubic grid, with the following construction (`cubic_scattering/`
1829 `graded_voxel`).

1830 *The voxel.* Each cell carries the nine-component state of §2, displacement
1831 and engineering strain, with the strain an *independent* unknown and not the
1832 gradient of the displacement: a strain derived from a cell’s projected dis-
1833 placement jumps across the faces between cells and leaves a spurious source
1834 there (§8). The state is expanded in the Legendre products of degree at

1835 most p in the cell's local coordinates $\boldsymbol{\xi} \in [-1, 1]^3$, and the contrast opera-
 1836 tor $\boldsymbol{\Delta}$ in those of degree at most r , its coefficients the L^2 projection of the
 1837 medium; $p = r = 1$ gives the first-moment voxel of 36 weights. The source
 1838 $\boldsymbol{\Delta}\psi$ is then a polynomial of degree $p + r$, expanded in monomials m_c . The
 1839 Lippmann–Schwinger equation is tested with each basis function over each
 1840 cell (Galerkin), and cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\boldsymbol{\xi}) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\boldsymbol{\xi}') d^3x d^3x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (\text{E.1})$$

1841 with $\mathbf{P}$ the 9×9 point propagator of §2.

1842 *One integral in place of two.* With $\mathbf{s} = \mathbf{u} - \mathbf{u}'$ the difference of the two
 1843 local positions, Eq. (E.1) becomes $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s$ over
 1844 $\mathbf{s} \in [-2h, 2h]^3$, where W_{ac} is a product over the three axes of one-dimensional
 1845 cross-correlations of the test and source polynomials: continuous, polynomial
 1846 on $[-2h, 0]$ and $[0, 2h]$, with kinks at 0 and $\pm 2h$. For cells that do not touch,
 1847 the kernel is smooth on each of the eight pieces, and a three-dimensional
 1848 Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev
 1849 distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional
 1850 quadrature of Eq. (E.1) to 10^{-9} one cell beyond contact.

1851 *The self cell and its 26 neighbours.* The static part of $\mathbf{P}$ is a sum of derivatives
 1852 of r^{-1} and r , distributions at the origin. Derivatives are moved onto W until
 1853 the kernel is at most as singular as $1/r$, and the kinks of W then carry
 1854 plane δ functions; every piece is a box or a rectangle with its corners on the
 1855 breakpoints, so the singular point is either a vertex of a piece, integrated
 1856 with Duffy's pyramids, or at least $2h$ away. The δ function of the inclusion
 1857 problem is carried by those plane δ functions and is not added by hand.

1858 The construction reproduces the sum rule $\sum_p \int_V \int_V \partial_p \partial_p |\mathbf{x} - \mathbf{x}'|^{-1} d^3x d^3x' =$
1859 $-4\pi V$, the biharmonic rule for r and the cube's Coulomb self-energy to 10^{-11} ,
1860 and every block equals the sum of the 64 blocks between the two cells' half-
1861 size sub-cells, with the polynomials re-expressed on them, to 2×10^{-15} . The
1862 dynamic remainder of $\mathbf{P}$ is at most weakly singular and is integrated by the
1863 Gauss rule of the pieces.

1864 *Symmetry, and the solve.* A block depends only on the offset $\mathbf{o}$ between the
1865 two cells. The background is isotropic and the cube is mapped to itself
1866 by its 48 signed permutations $\mathbf{Q}$, under which $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^\top$, with
1867 $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$ and $\mathbf{R}_6$ the representation of $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^\top$ on Voigt
1868 vectors. With the test and source polynomials transforming by matrices $\mathbf{T}$
1869 and $\mathbf{U}$, $\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^\top$, so one block is computed for each
1870 orbit of offsets and the rest follow exactly. The coupling is then a convolution
1871 over the grid, applied by the fast Fourier transform on a $(2n-1)^3$ circular
1872 embedding, and the system is solved by GMRES preconditioned with each
1873 cell's own block. Where both can be run, the solution equals that of a dense
1874 solve to 10^{-10} relative (`tests/test_graded_voxel_fft.py`). Every cell that
1875 overlaps the body is kept, each with the L^2 projection of its contrast by a
1876 six-point Gauss rule per axis; keeping only the cells whose centre lies inside
1877 the body drops the contrast of the partly filled cells, and the mean-only
1878 voxel then converges at order one instead of two. The far field is read at
1879 5×10^8 radii, where the near-field terms that the asymptotic form omits are
1880 negligible.

1881 *The first-moment voxel on the graded sphere.* The voxel with $p = r = 1$ is
1882 applied to the graded sphere of §10. To isolate the discretisation error from

1883 the rescattering the grid cannot yet resolve, the contrast is first scaled by
 1884 10^{-6} , and the exact sphere is solved at the same scaled contrast, so that no
 1885 Born approximation enters the reference: the relative error is then that of
 1886 the scheme's first-order term, in which the field is the incident one carried
 1887 across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the
 1888 error is 7.5×10^{-5} , 5.0×10^{-6} and 1.6×10^{-6} at $k_S a = 0.5$, and 2.3×10^{-4} ,
 1889 9.1×10^{-5} and 2.8×10^{-5} at $k_S a = 1$. The apparent orders between 6 and
 1890 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a
 1891 voxel at $n = 4$ being as wide as the graded shell (`scripts/pilot_graded_`
 1892 `voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16
 1893 voxels across, where the error is 4.9×10^{-6} at $k_S a = 0.5$ and 5.9×10^{-6} at
 1894 $k_S a = 1$, and the apparent order approaches four at a rate set by the number
 1895 of voxels across the graded shell.

1896 **Appendix F. The uniform-strain closure against the exact sphere**

1897 *Against the exact sphere.* These statements are tested against the exact elas-
 1898 tic Mie solution for a sphere, with no Eshelby formula in between. The
 1899 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 1900 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 1901 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 1902 route of Eq. (26) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 1903 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 1904 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 1905 isotropic shear average, Eq. (32), is the depolarisation read from Mie's E_2
 1906 to 10^{-8} . The same moment taken as a principal value, over the ball with

1907 a small sphere excluded around the origin, has zero depolarisation in every
 1908 channel. The whole static response of the sphere is the δ term.

1909 At finite frequency the closure uses the dynamic ball moments, which are
 1910 in closed form. It departs from Mie at second order in ka and first order in
 1911 the contrast, and the departure is closed-form too: the ball's moments are
 1912 elementary, and Mie's low-frequency series with the contrast kept exact gives
 1913 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (\text{F.1})$$

1914 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 1915 the radiation damping, agree exactly in all three channels. So the closure's
 1916 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 1917 own. That is the order the hierarchy assigns to the omitted second gradi-
 1918 ents (§6.3), and it has the same origin, the variation of the field across the
 1919 scatterer, as the collocation error of §7. Eq. (F.1) is derived symbolically
 1920 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 1921 the numerical Mie solve and an independent implementation of the closure
 1922 to 5×10^{-4} , which is the size of the next order in the contrast.

1923 Appendix G. The scattering series of the layer, and the projection- 1924 error law

1925 *The scattering series.* Multiply the whole contrast, $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$
 1926 together, by a number s , and let $u(s)$ be the scattered displacement at an

observer at depth z_o outside the layer. It vanishes at $s = 0$ and is analytic in s there, and its expansion is the Born series,

$$u(s) = s T_1 + s^2 T_2 + s^3 T_3 + \dots, \quad (\text{G.1})$$

so that T_n is the part of the field at the full contrast, $s = 1$, that has been scattered n times.

The first two terms of the exact layer. They are integrals over the layer. With the incident state $w^0 = (u^0, \varepsilon^0)$, the contrast $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ with profile $f(z)$, and the kernel $\mathbf{K}$ of Eq. (38), whose first row $(g, g')(z_o - z)$ gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (\text{G.2})$$

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (\text{G.3})$$

Here w^1 is the state scattered once, inside the layer; its strain includes the local term of the kernel, $-f(z) \Delta M \varepsilon^0(z)/M_P$. Each further term repeats the step, $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$. For a plane force at z_s above the layer, $w^0(z) = c e^{ik(z-z_s)}(1, ik)$ with $c = i/(2M_P k)$, every integrand is an exponential times the profile. Writing $a = \omega^2 \Delta \rho$ and $b = \Delta M$, and carrying out the products of Eqs. (G.2) and (G.3), the terms seen in reflection ($z_o < 0$) and in transmission ($z_o > D$) are, for any profile,

$$T_1^R = c^2 e^{-ik(z_o + z_s)} (a + bk^2) \int_0^D f(z) e^{2ikz} dz, \quad T_1^T = c^2 e^{ik(z_o - z_s)} (a - bk^2) \int_0^D f(z) dz, \quad (\text{G.4})$$

1943

$$\begin{aligned}
T_2^R &= c^2 e^{-ik(z_o+z_s)} \left[c(a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 e^{2ikz} dz \right], \\
T_2^T &= c^2 e^{ik(z_o-z_s)} \left[c(a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c(a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\
&\quad \left. + \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 dz \right].
\end{aligned}
\tag{G.5}$$

1944 The first-order reflection is the Fourier transform of the profile at the wavenum-
1945 ber $2k$, and the first-order transmission its integral. In each second-order
1946 term the last integral is the local term of the kernel, in which both scat-
1947 terings are at one point and the profile appears squared; the others are the
1948 two scatterings at different depths. In transmission the wave scattered twice
1949 forwards carries the contrast of velocity twice, $(a - bk^2)^2$, and the wave scat-
1950 tered backwards and then forwards again the contrast of impedance twice,
1951 $(a + bk^2)^2$. For the uniform layer ($f = 1$) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a + bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a - bk^2) D,
\tag{G.6}$$

1952

$$\begin{aligned}
T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_P} \left[-i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\
T_2^T &= \frac{i c^2 e^{ik(z_o-z_s)}}{8k^3 M_P} \left[a^2(1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2(e^{2ikD} - 1 + 2kD(kD - i)) \right. \\
&\quad \left. + b^2 k^4(1 - e^{2ikD} + 2kD(kD - 3i)) \right].
\end{aligned}
\tag{G.7}$$

1953 The reflection sees the contrast of impedance, $a + bk^2 = k^2 M_P(\Delta\rho/\rho +$
1954 $\Delta M/M_P)$, and the transmission that of velocity, $a - bk^2$. As $kD \rightarrow 0$, $T_2^R \rightarrow$
1955 $-c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_P$: the local term alone, which is the starting point of
1956 the argument at the end of this appendix. For the smooth profile the integrals

of Eqs. (G.4) and (G.5) are evaluated in closed form symbolically, and by quadrature they return the same values to 10^{-16} . Equations (G.6) and (G.7) equal, identically, the first two Taylor coefficients in s of the exact reflection and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At $\omega = 300 \text{ rad s}^{-1}$, $|T_2|/|T_1|$ is 0.039 in reflection and 0.17 in transmission for the uniform layer.

The first two terms of the scheme. Write Eq. (38) as $(\mathbf{M} - \mathbf{K}) \mathbf{c} = \mathbf{b}$, with $\mathbf{M}$ the diagonal matrix of the $\int_i \phi_a^2 dz$, $\mathbf{K}$ the matrix of the coupling blocks times Δ_1 , and $\mathbf{b}$ the moments of the incident field, and let $\mathbf{r}$ be the row that gives the scattered displacement at the observer from the coefficients, $u = \mathbf{r} \mathbf{c}$. Scaling the contrast by s scales $\mathbf{K}$ and $\mathbf{r}$ by s , so $u(s) = s \mathbf{r} (\mathbf{M} - s \mathbf{K})^{-1} \mathbf{b}$ and

$$T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}, \quad T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}. \quad (\text{G.8})$$

$\mathbf{M}^{-1} \mathbf{b}$ is the projection of the incident field onto each cell's functions: the first term needs no solve. Splitting $\mathbf{K}$ into its blocks within a cell and its blocks between cells splits T_2^{scheme} into the part scattered twice in one cell and the part scattered in two; the same split of the inner integral of Eq. (G.3) splits the exact T_2 .

A check by differences. The terms can also be read from solutions at four scaled contrasts, $s = \pm\epsilon$ and $\pm 2\epsilon$, with no knowledge of the formulas above. The odd and even parts of u in s separate the odd orders from the even, and the two step sizes remove the next term of each (the five-point central differences for $u'(0)$ and $\frac{1}{2}u''(0)$):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (\text{G.9})$$

1978 A negative s is a medium slower and lighter than the background by the
1979 same fractions, as legitimate a medium as the other. With $\epsilon = 0.01$ the
1980 remainders $4\epsilon^4 T_5/T_1$ and $4\epsilon^4 T_6/T_2$ are below 2×10^{-12} for this layer, and the
1981 formulas applied to the exact layer return Eqs. (G.6) and (G.7) less exactly
1982 those remainders, to 10^{-17} . The measurements of §9 use the differences for
1983 the scheme and the exact layer alike, each in its own arithmetic, and they
1984 agree with Eqs. (G.2), (G.3) and (G.8).

1985 *Why the terms must be separated.* The law of Eq. (44) is a statement
1986 about T_2 . At the full contrast the terms of third and higher order are 7% of
1987 the second and carry errors of their own, and if every order above the first
1988 is taken together the ratio of Eq. (44) reads 0.90 to 0.94 where the second-
1989 order term alone gives 1.00 to 1.01. The shortfall is itself the single site's. In
1990 the long-wave limit the kernel is local, the exact strain is $\varepsilon_0/(1 - cf)$ with
1991 $c = -\Delta M/M_P$, and the scheme's is the Galerkin solution of $\varepsilon = \varepsilon_0 + cf\varepsilon$ in
1992 each cell on that cell's polynomials; the relative error of the term of order
1993 n is $(\int f^n dz - \int (\Pi f)^n dz)/\int f^n dz$, 2.455 times $\mathcal{E}_q$ for the third term of
1994 this profile, which has the opposite sign to the second. At $\omega = 30 \text{ rad s}^{-1}$
1995 the whole nonlinear part has 0.896 times $\mathcal{E}_q$, and the cell-by-cell Galerkin
1996 solution reproduces it to four digits with c from the contrast (`scripts/`
1997 `measure_layer_t3_low.py`).

1998 Equation (44) is exact in the long-wave limit, for a reason that shows what
1999 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
2000 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
2001 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
2002 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-

2003 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order
 2004 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f \, dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) \, dz = \|\Pi_q f\|^2, \quad (\text{G.10})$$

2005 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 2006 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 2007 scattering takes place in the cell of the first: T_2 is the second-order term of
 2008 the single-site T -matrix, and Eq. (44) is a statement about the single site.
 2009 The T -matrix sums the rescattering within a cell exactly for the cell as it
 2010 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 2011 cell is not the medium. The split is measured directly (`scripts/measure_`
 2012 `layer_bases.py`): at $\omega = 300 \text{ rad s}^{-1}$ on 4, 8 and 16 cells, T_2 of the scheme
 2013 and of the exact layer are each divided into the part whose two scatterings
 2014 lie in one cell and the part between different cells. For all nine pairs the
 2015 error of the same-cell part is 0.998 to 1.013 times $|T_2| \mathcal{E}_{\min(p, r)}$, and that of
 2016 the part between cells at most 0.002 times it. The coupling between cells
 2017 contributes only the departure from the limit. Measured, the ratio of the
 2018 two sides of Eq. (44) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} ,
 2019 4.3×10^{-3} , 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 :
 2020 as $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (G.10)
 2021 is verified symbolically for all nine pairs, and the projection errors are com-
 2022 puted exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each
 2023 halving of the cells.

2024 **Appendix H. Relation to other methods**

2025 *Relation to the coupled-dipole method.* The same partition appears in the
2026 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
2027 relation, the lattice dispersion relation [11], the polarisability taken from the
2028 exact static solution [12] and the integrated Green tensor [13] all adjust the
2029 single site so that the lattice reproduces the continuum. For space-filling
2030 cubes the adjustment is neither needed nor a free parameter: the tiling iden-
2031 tity fixes the local field exactly, the self term of the collocation closure is the
2032 one that makes it cancel, and what is left is a closed-form error rather than
2033 an empirical one. That the method is a collocation of the volume integral
2034 equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (35)
2035 is the same statement for the elastic lattice with the self term evaluated over
2036 the finite cube, and it is what makes the error a midpoint-rule error. The
2037 filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-
2038 limit the source to the grid; the source-cell average does the complementary
2039 thing exactly, its form factor vanishing on the reciprocal lattice, and the
2040 spectral construction of Eq. (B.1) is the periodic coupled-dipole method of
2041 Chaumet et al. [16] with the Galerkin double average in place of the point
2042 dipole. The second-order convergence reported for the DDA by Yurkin et al.
2043 [3] corresponds to the uniform-field voxel’s, and the opposite-signed con-
2044 stants of Eq. (36) explain why a polarisability correction can improve one
2045 observable only at the expense of another, as the package’s own form-factor
2046 corrections do in reflection and transmission. Going beyond second order
2047 requires a richer internal field, which is the point of higher-order bases for
2048 volume integral equations [70, 2]; the first-moment voxel is the lowest such

basis, and in its spectral form the Galerkin double average makes the lattice sum converge without Ewald splitting.

Relation to the spectral element method. The voxel of degree p is close kin to the spectral element [71], which was taken up for seismic wave propagation in the 1990s, is set out for three-dimensional models by Komatitsch and Tromp [72], and is now a standard tool for it. Both divide the medium into hexahedra, represent the field in each by polynomials of high degree, and determine the coefficients by a Galerkin condition on a weak form. The spectral element uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of the element; the voxel uses the Legendre polynomials themselves. Four of the present results have their counterparts there.

1. *A diagonal mass matrix.* The spectral element obtains one by integrating at the Gauss–Lobatto–Legendre points, at which each interpolant vanishes but one [72]. That rule is not exact for the product of two interpolants, so that the diagonal form is reached, in their words, by a process of subintegration, as in the finite-element methods that lump the mass. Here the Gram matrix $\mathbf{M}$ of Eq. (G.8) is diagonal with no approximation, because the Legendre functions are orthogonal on the cell; this is why the first-order term needs no solve. The two routes serve different ends: theirs an explicit march in time, ours the terms of the scattering series.

2. *The doubled order.* A scattered amplitude is a functional of the field, and a Galerkin method gives a functional to twice the order of the field itself. The order $2p + 2$ is this doubling: the leading error of Eq. (39) is the product of the part of the incident wave that the cell’s polynomials

miss and the part of the outgoing wave that they miss, each of order h^{p+1} .

3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [72]. The rule $\min(2p+2, 2r+2)$ says why that is the right choice, and what a coarser representation of the medium would cost.

4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [72]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [72]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a cou-

2099 pling between every pair of voxels, and a background restricted to one whose
 2100 Green's tensor is known. It differs also from the boundary integral methods,
 2101 which are restricted to a finite number of homogeneous regions [72]: a vol-
 2102 ume integral equation admits a medium that varies within the scatterer. The
 2103 voxel scheme is therefore a candidate where the heterogeneity is localised in
 2104 a layered medium, and not a replacement for the spectral element in a model
 2105 that is heterogeneous throughout. A spectral-element mesh is designed by a
 2106 rule of resolution, about five grid points to the shortest wavelength at poly-
 2107 nomial degrees between five and ten [72]. The error analysis of this paper is
 2108 the counterpart of that rule for the integral formulation: it gives the size and
 2109 the degree of the voxels for a required accuracy in closed form, and it adds a
 2110 term, the projection error of the medium.

2111 **Appendix I. The cost of higher order, and what the error analysis** 2112 **offers a fast solver**

2113 *The cost of fourth order.* A first-moment voxel carries four times as many
 2114 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 2115 forms show what that buys. To reach an error ε in reflection, the uniform
 2116 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 2117 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 2118 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 2119 number of unknowns by about 5×10^4 (these are the Born-order constants;
 2120 at the contrast studied the first-moment voxel's measured errors lie within
 2121 5% of them, while the uniform voxel's transmission constant grows with the
 2122 contrast, so the comparison is if anything conservative). The stratified model

2123 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 2124 moment voxels for the same accuracy. Higher moments continue the trend
 2125 in one dimension: by Eq. (40) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 2126 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 2127 a quarter of a wavelength. In three dimensions, with the Legendre products
 2128 of total degree at most p , a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per
 2129 state component, against 4 for the first-moment voxel. The voxel of degree
 2130 2 converges at sixth order in the layer at oblique incidence (Table B.15);
 2131 that of degree 3 has not been built in three dimensions. The gain is largest
 2132 precisely where it matters: when the model varies on scales that are coarse
 2133 compared with the accuracy the wavefield must have.

2134 *What the error analysis offers a fast solver.* A voxel scheme is dense, and
 2135 for a large medium it is solved iteratively with a fast product. The adaptive
 2136 integral method [73] and the precorrected fast Fourier transform [74] are
 2137 the standard forms: the sources are projected onto a regular grid, in the
 2138 first by matching their multipole moments and in the second by matching
 2139 the potentials they produce at a set of test points, the interactions between
 2140 distant cells are evaluated on that grid by fast Fourier transforms, and those
 2141 between near cells are computed directly and corrected. Such a method has
 2142 three accuracies to set, that of the discretisation, that of the projection, and
 2143 the radius of the near zone, and they are usually set by trial. The results of
 2144 this paper set them beforehand, in five ways.

- 2145 1. *The target is known.* The discretisation error is known before the solve,
 2146 in order and in size: the first-order factor of a cell, Eq. (39) and its
 2147 form for any pair of wavenumbers (§9), and the projection error of the

2148 medium, Eq. (44). A fast product need be no more accurate than that,
2149 and it is known when it is accurate enough.

2150 2. *The two errors do not mix.* The error of the scheme is the single site's:
2151 the coupling between cells, averaged over the cells, is exact (§5). An
2152 approximation of the far coupling to a set tolerance therefore adds its
2153 own error to the discretisation's and does not alter it, so the two can
2154 be budgeted separately.

2155 3. *For cubes on a lattice there is nothing to project.* The cells are the reg-
2156 ular grid. The coupling depends on the separation of the cells alone, so
2157 the product is a convolution, block by block in the polynomial degrees,
2158 and the fast Fourier transform applies to it exactly, as in Georgakis
2159 et al. [2] for the linear voxel. What the projection onto the grid ap-
2160 proximates in general is here an identity.

2161 4. *For cells of several sizes the moments are already in hand.* Where the
2162 cells differ in size or degree and are mapped to a uniform auxiliary
2163 grid, the quantities to be matched are the polynomial moments of a
2164 cell, which are the unknowns of the scheme; and the first-order factor
2165 says how many of them matter at a given kd .

2166 5. *A stratified background splits as the method requires.* The Green's ten-
2167 sor of a plane-stratified background is the whole-space tensor, which is
2168 invariant under translation in three dimensions and carries all the sin-
2169 gularity, plus a reverberation, Eq. (C.1), which is smooth, is invariant
2170 under lateral translation, and separates in depth into one-dimensional
2171 moments. The first is the convolution above; the second is a fast Fourier
2172 transform laterally and a product in depth, with no near-zone correc-

tion except where a voxel touches an interface. A splitting of the kernel into a term that carries the singularity and a nonsingular remainder is used to a related end by Sharma and Triverio [75], for surface integral equations in homogeneous and in layered backgrounds: there the singular term is independent of frequency, so that the direct near-zone integrals are computed once for a whole sweep in frequency. And because the orders and the error law are unchanged by the background (§9), the error budget of the whole space carries over to it.

No such solver is built here. These are consequences of the analysis, and they are to be tested.

Appendix J. The evidence for each result, and where to reproduce it

Table J.22 lists each result with the evidence for it. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, GradedContrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the moment integrals are `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python half of the last in `scripts/gate_cube_shear_split.py`; the stratified reference against Kennett’s recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_`

2197 `vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded,
 2198 with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded
 2199 sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with
 2200 the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_
 2201 sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_
 2202 vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.
 2203 wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the
 2204 independent implementation of the first-moment voxel is `scripts/crosscheck_
 2205 first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_
 2206 second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_
 2207 graded_contrast.py`; the projection-error law and its split are measured by
 2208 `scripts/measure_layer_bases.py`, in a stratified background by `scripts/
 2209 measure_layer_stratified_background.py`, `scripts/measure_oblique_
 2210 stratified_background.py` and `scripts/measure_oblique_two_term.py`,
 2211 and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_
 2212 degree.py`; the convergence figure is drawn by `scripts/plot_convergence_
 2213 orders.py`; the near field of the sphere is measured by `scripts/measure_
 2214 graded_sphere_near_field.py`; the moments of every grade are in `Mathematica/
 2215 CubeMomentHigherGrades.wl`, `Mathematica/CubeMomentStore.wl` and `Mathematica/
 2216 CubeMomentCanonical.wl`, with their independent evaluation in `scripts/
 2217 crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is
 2218 `Mathematica/ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_
 2219 layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_
 2220 ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.
 2221 py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All

²²²² are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

Table J.22: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (26)	sum rule (27) exact; M, Q equal an independent archive, 81/81
Channels (29)–(31)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (32)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (F.1)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (35)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (36)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (38)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (B.1)	order 4, P and S, 20° and 40°

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2225 the Research School of Earth Sciences of the Australian National University,
2226 Canberra [21].

2227 **Declaration of competing interest**

2228 The author declares no known competing financial interests or personal
2229 relationships that could have appeared to influence the work reported in this
2230 paper.

2231 **Declaration of generative AI and AI-assisted technologies in the** 2232 **manuscript preparation process**

2233 *Software development..* Software development followed a human-directed,
2234 AI-assisted workflow. The author designed the software originally, but only
2235 directed and validated recent improvements, which build on the author’s
2236 original Fortran implementation (the `PhD_fortran_code` folder of the repos-
2237 itory given under Data availability) and an extensive library of Mathematica
2238 notebooks, developed and extended since 1992, that carry out and verify the
2239 symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus
2240 5.5) operated under human-in-the-loop supervision to implement specified
2241 changes and diagnostic tests. The design, the choice of numerical methods,
2242 and the resolution of the principal technical difficulties were the author’s. All
2243 AI-generated changes were subject to code review by the author and validated
2244 by automated tests and independent physical checks such as reciprocity, op-
2245 tical theorem, Hamiltonian symmetry, analytic limits and convergence.

2246 *Manuscript preparation..* During the preparation of this work, the author
2247 used Claude (Anthropic) in order to search for relevant open-access research,
2248 to write the TikZ code of the diagrams and to help redraft the paper into more
2249 coherent sections for readability. After using this tool, the author reviewed
2250 and edited the content as needed and takes full responsibility for the content
2251 of the published article.

2252 **Data availability**

2253 The Mathematica notebooks and Python scripts that derive and check
2254 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
2255 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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